

Cisco Firepower Next-Generation Firewall VPN Lab – FMC v1.6

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IMPORTANT! This content is community developed and is not subject to standard dCloud verification or support. Please contact dCloud Support for more information.

About This Guide

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Requirements

The table below outlines the requirements for this preconfigured demonstration.

Table 1. Requirements

Required	Optional
Laptop	Cisco AnyConnect®

About this Solution

IT teams have been asked to manage security using a patchwork of siloed point products, starting with legacy next-generation firewalls (NGFW), which were created with a focus on applications and bolted on best effort threat protection. As such, these legacy NGFWs are unable to provide an enterprise with the contextual information, automation, and prioritization required to handle today's modern threats.

Cisco Firepower is an integrated suite of network security and traffic management products, deployed either on purpose-built platforms or as a software solution. The system is designed to help you handle network traffic in a way that complies with your organization's security policy - your guidelines for protecting your network.

This allows the Cisco Firepower NGFW to evolve with a focus on enabling enterprises to stop, prioritize, understand, and automate responses to modern threats in real-time. Firepower NGFW is unique in its threat-focus, with a foundation of comprehensive network visibility, best-of-breed threat intelligence and highly effective threat prevention to address both known and unknown threats. Firepower NGFW also enables retrospective security, through Advanced Malware Protection, that can "go back in time" to quickly find and remediate sophisticated attacks that may have slipped through defenses leading to a significant reduction in time-to-detection (TTD) for Cisco customers compared to industry averages.

In this lab you will leverage the FMC to provision secure access for remote corporate users and configure secure connectivity between Main/HQ and Branch/Remote office sites. Using FMC, you will configure a Site to Site VPN between a Main Office and a Branch office location. Also, you will use FMC to provision Remote Corporate Users to connect to the Main Office and Branch location. The lab will showcase the various steps involved and new features available such as MFA and LDAP attributes.

Topology

This content includes preconfigured users and components to illustrate the scripted scenarios and features of the solution. Most components are fully configurable with predefined administrative user accounts. You can see the IP address and user account credentials to use to access a component by clicking the component icon in the **Topology** menu of your active dCloud session and in the scenario steps that require their use.

NOTE: For simplicity, not all IP addresses and VLANs are shown.

Login credentials are provided in line in the guide steps as needed. However, for your convenience, the following table lists the credentials used in these scenarios.

VM	Login	Password
FMC	admin	C1sco12345
All NGFWs	admin	C1sco12345
All Windows Workstations	Administrator	C1sco12345
Splunk	admin	C1sco12345
ISE and ISE PIC	admin	C1sco12345
Inside Linux Server	root	C1sco12345
Outside Linux Server	root	C1sco12345
CSR	admin	C1sco12345

Table 3. User Credentials

Login	Password	Details
dilbert	C1sco12345	Active Directory user in Engineering group
harry	C1sco12345	Active Directory user in HR group
ira	C1sco12345	Active Directory user in Finance and Investment groups
rita	C1sco12345	Active Directory user in IT group
alicia	C1sco12345	ISE local user, FDM Admin user
oliver	C1sco12345	ISE local user, FDM Read Only user
victoria	C1sco12345	ISE local user, VPN user
william	C1sco12345	ISE local user, FDM Read Write user

Get Started

BEFORE PRESENTING

Cisco dCloud strongly recommends that you perform the tasks in this document with an active session before presenting in front of a live audience. This will allow you to become familiar with the structure of the document and content.

It may be necessary to schedule a new session after following this guide in order to reset the environment to its original configuration.

PREPARATION IS KEY TO A SUCCESSFUL PRESENTATION.

Follow the steps to schedule a session of the content and configure your presentation environment.

1. Initiate your dCloud session. [\[Show Me How\]](#)

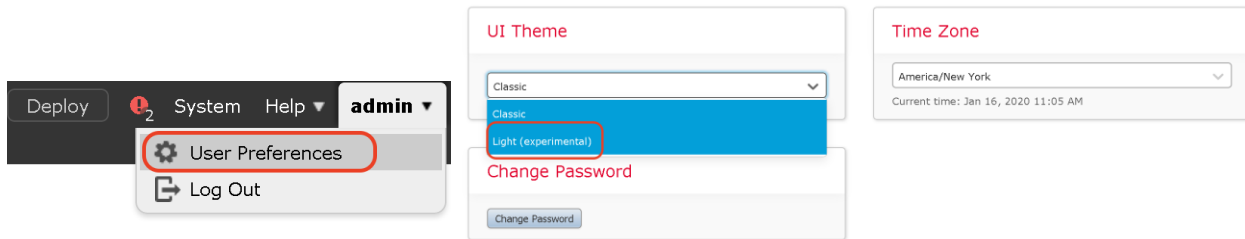
NOTE: It may take up to 10 minutes for your session to become active.

2. It is assumed in this guide that you are connecting to all devices from the Jumpbox RDP session, using the provided details.

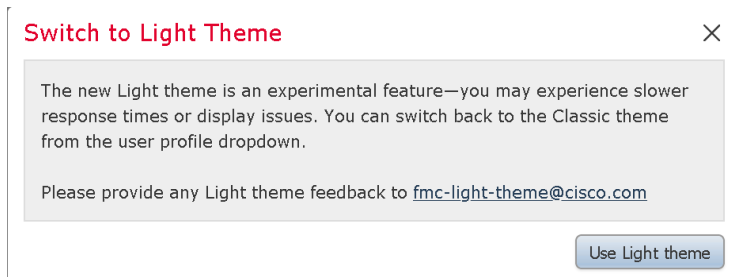
NOTE: You can also connect to the workstation with **Cisco AnyConnect VPN** [\[Show Me How\]](#) and the **local RDP client on your laptop** [\[Show Me How\]](#)

Jumpbox: **198.18.133.50**, Username: **administrator**, Password: **C1sco12345**

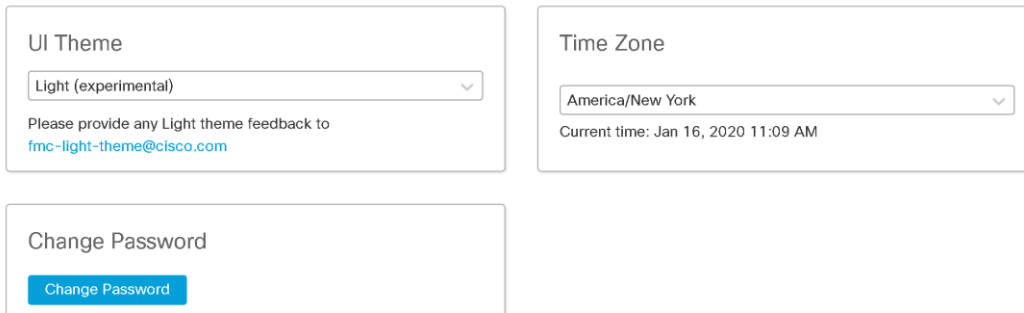
3. From the jumpbox, open **Firefox**. It should open with FMC login page as the default page. The login name and password will be prepopulated. Click **Login**
4. Navigate to **admin > User Preferences** in the top right-hand corner. Under the **General** tab, click on the dropdown under **UI Theme** and select **Light** theme.



5. A popup will show up with a disclaimer about the Light Theme. Click on **Use Light theme**.



6. The FMC UI will change from the classic UI to new UI.



The above step is optional for Firepower release 6.5. In Firepower release 6.6, the above UI theme is used default. However, all the lab screenshots were captured using the new UI theme. It has no impact on the configuration steps.

NOTE: You can also connect to the workstation using the Cisco dCloud Remote Desktop client [\[Show Me How\]](#). The dCloud Remote Desktop client works best for accessing an active session with minimal interaction. However, many users experience connection and performance issues with this method.

Scenario 1. Securing Remote Users for Corporate Access

With some of the new functionality being added on the FMC in 6.4, it will now be possible to setup an effective and secure remote access VPN solution using Anyconnect Client. Multi-factor authentication (MFA) and ISE posture assessment coupled with next gen firewall features allow us to gain a deeper visibility into remote VPN users and enforce Malware, File and IPS policies. To allow for MFA, we will be using a combination of Client certificate + RADIUS Credentials (ISE integrated with AD) + Duo OTP using the Push method. It is possible to use MFA in conjunction with Posture Assessment due to the introduction of the Secondary Authentication Server or Double Authentication.

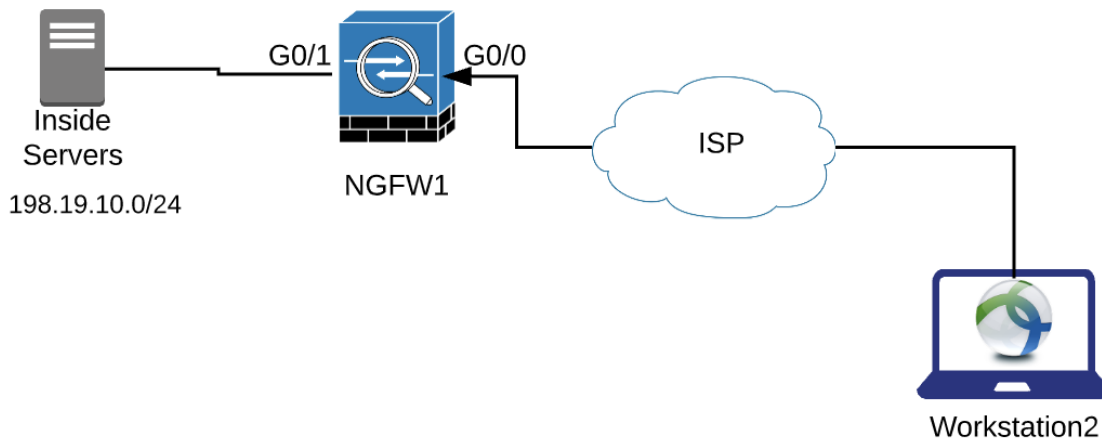
NOTE: In case the HA exercise was done before this lab, please select HA_TEST for devices where prompted, in place of NGFW1. Also, in order to save time, ISE is pre-configured with all required configuration for all the lab exercises. This includes the selection of group policy and IP pool based on AD group membership. **Because of this, the name of the new group policy and IP pool must be exactly the names given in the instructions.** If you want to review the ISE configuration, see Appendix D.

The exercise consists of the following configuration tasks:

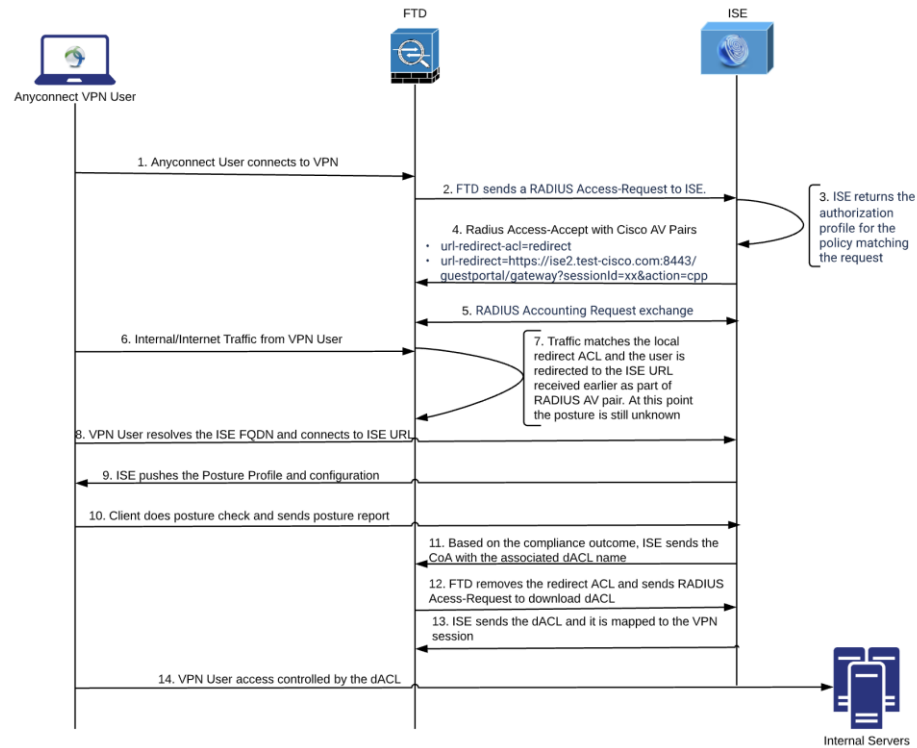
- Run RA VPN Wizard
- Configure DNS settings
- Modify the Access control and NAT policies
- Run the RA VPN setup wizard
- Deploy and test the configuration

The objectives of this scenario are:

- To configure an AnyConnect VPN client to connect to the NGFW
- To explore various authentication methods for Anyconnect VPN users.
- To test the Intrusion prevention and Malware configuration of the NGFW for the Remote Access VPN users.



In this exercise, we will use ISE to validate the posture of the user machine. The flow of events will be as below :



NOTE: The FTD should have an ACL configured with the exact name '**redirect**', matching the name of the ACL returned in the AV pair attribute returned by ISE. The ACL name is case-sensitive. For more details of the ISE posture workflow, please take a look at the [ISE posture flow guide](#).

Pre-Configuration Steps

Apply Licenses needed for this Scenario

1. Go to System>Licenses>Smart Licenses
 - a. Click on Edit Licenses
 - b. Select AnyConnect Apex
 - c. Select **NGFW1**
 - d. Click **Add** and then **Apply**

Verify objects needed for this scenario

1. In the FMC, navigate to **Objects > Object Management**. Click **Network**

Firepower Management Center

Overview Analysis Policies Devices Objects AMP Intelligence

Object Management

Intrusion Rules

Add Network

Filter

Name	Value	Type	Override
any	0.0.0.0/0	Group	
any-ipv4	0.0.0.0/0	Network	
any-ipv6	::/0	Host	
DNS_Server	198.19.10.100	Host	
Extranet129	198.18.129.0/24	Network	
Extranet130	198.18.130.0/24	Network	
Extranets	Extranet129 Extranet130	Group	
Infrastructure	198.19.10.80-198.19.10.130	Range	

2. Using the **Filter** search box, you can review the objects that were created in advance. Enter the below names to verify that the objects have been created as below:

- a. **VPNPoolIPs** - You can click on the object name to confirm that it contains IP address range 198.19.10.57-198.19.10.62. This object will be used in NAT-exemption and represents the IP addresses to be assigned to the Anyconnect VPN user. The following screen captures shows how the object was configured.

Firepower Management Center

Overview Analysis Policies Devices Objects AMP Intelligence

VPNPoolIPs

Name	Value	Type	Override
VPNPoolIPs	198.19.10.57-198.10.10.62	Range	

- b. **LAN_Network** - You can click on the object name to confirm that it contains IP address network 198.19.10.0/24. This is the internal corporate network protected by the VPN headend. The following screen captures shows how the object was configured.

Network

LAN_Network

Name	Value	Type	Override
LAN_Network	198.19.10.0/24	Network	

- c. **ISE_Server** - You can click on the object name to confirm that it contains IP address 198.19.10.130. The following screen captures shows how the object was configured.

Network

ISE_Server

Name	Value	Type	Override
ISE_Server	198.19.10.130	Host	

- d. **DNS_Server** - You can click on the object name to confirm that it contains IP address 198.19.10.100. The following screen captures shows how the object was configured.

Network				
A network object represents one or more IP addresses. Network objects are used in various places, including access control policies, network variables, intrusion rules, identity rules, network discovery rules, event searches, reports, and so on.				
Name	Value	Type	Override	
DNS_Server	198.19.10.100	Host		

Create and configure a RADIUS server group for VPN user Authentication

A RADIUS Server Group object can contain multiple RADIUS Servers. Here, we will create a RADIUS Server Group with ISE as the RADIUS Server. It will be used for Authentication and Authorization of Remote Access VPN Users.

- Within the object management page, select **RADIUS Server Group** from the left navigation pane. Click **Add RADIUS Server Group**.
 - Call the group **ISE_RADIUS**.
 - Check the **Enable dynamic authorization** checkbox.
 - Click the **+** sign to add a **RADIUS server**.

The screenshot shows the 'Add RADIUS Server Group' configuration page. The left navigation pane has 'RADIUS Server Group' selected. The main form has the following fields and settings:

- Name:** ISE_RADIUS (1)
- Description:** (empty)
- Group Accounting Mode:** Single
- Retry Interval:** 10 (1-10) Seconds
- Realms:** (empty)
- Enable authorize only:** ☐
- Enable interim account update:** ☐
- Interval:** 2 (1-120) hours
- Enable dynamic authorization:** ☒ (2)
- Port:** 1700 (1024-65535)
- RADIUS Servers (Maximum 16 servers):** (3)

IP Address/Hostname
No records to display

The 'New RADIUS Server' dialog box is open with the following settings:

- IP Address/Hostname:** 198.19.10.130 (4)
- Authentication Port:** 1812 (1-65535)
- Key:** C1sco12345 (5)
- Confirm Key:** C1sco12345 (6)
- Accounting Port:** 1813 (1-65535)
- Timeout:** 10 (1-300) Seconds
- Connect using:** Specific Interface (7)
- Interface:** InZone1 (8)
- Redirect ACL:** redirect (9)

Buttons for 'Cancel' and 'Save' are at the bottom of the dialog (10).

- Enter the following information, leaving other attributes as default.
 - For IP Address/Hostname enter **198.19.10.130**.
 - For Key enter **C1sco12345**. Enter the same value for the **Confirm Key** field.
 - Select the **Specific interface** radio button and select **InZone1** from the drop-down menu.
 - click the **+** sign next to **Redirect ACL** to add an extended ACL to be used as ISE redirect ACL.

Note: Create an extended access list called **redirect** that looks like the below screenshot. This will be used to determine what traffic is to be redirected to ISE while posture assessment is taking place. The ACEs where the action is block will be exempted

from redirection. Some of the traffic such as DNS and ISE communication has to be exempted from redirection since we want the Anyconnect Client to be able to resolve the ISE redirect URL and communicate with ISE for posture config download (explained in the flow diagram earlier).

- i. Type in the name **redirect**
- ii. Click **Add** under the Entries to add the below 3 access-list entries:
 1. Select Action as **Block**, Destination as **ISE_Server** and Destination port as **TCP 8443**. Click **Add**
 2. Section Action as **Block**, Destination as **DNS_Server** and Destination port as **DNS_over_UDP**. Click **Add**
 3. Select Action as **Allow**, and rest as defaults. Click **Add**
- iii. Click **Save** after adding all the entries.

The screenshot shows the 'New Extended Access List Object' dialog box in the Cisco dCloud configuration interface. The 'Name' field is set to 'redirect'. An 'Add' button is located to the right of the 'Entries (0)' section. Below this is a table with the following columns: Sequence, Action, Source, Source Port, Destination, and Destination Port. The table is currently empty, displaying 'No records to display'. At the bottom of the dialog box are 'Cancel' and 'Save' buttons. In the background, the 'Redirect ACL' dropdown menu is visible with a '+' button next to it.

Add Extended Access List Entry

Action: ☐ Block

Logging: ☐ Default

Log Level: ☐ Informational

Log Interval: 300 Sec.

Network **Port**

Available Ports

- AOL
- Bittorrent
- DNS_over_TCP
- DNS_over_UDP
- FTP
- HTTP
- HTTPS
- IMAP

Selected Source Ports (0)

any

Selected Destination Ports (1)

TCP (6):8443

Protocol TCP (6) Port

Note: Be mindful of the name of the access-list. It should be 'redirect'. The two screenshots above the steps of adding 1st access-list entry. The 2nd and 3rd access-list entries can be added in a similar fashion.

New Extended Access List Object

Name:

Entries (3)

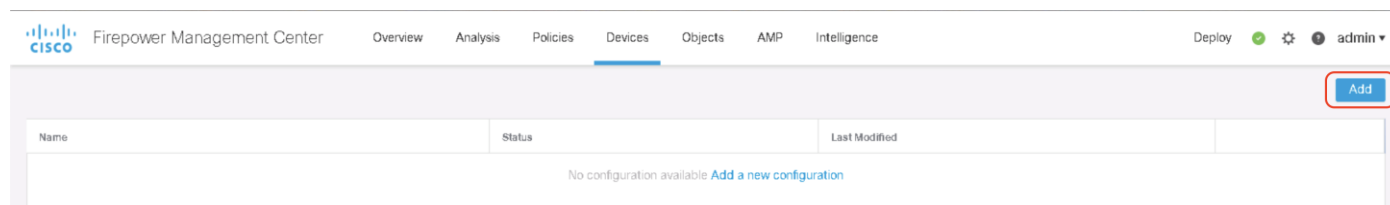
Sequence	Action	Source	Source Port	Destination	Destination Port	
1	Block	Any	Any	ISE_Server	TCP (6):8443	Edit Delete
2	Block	Any	Any	DNS_Server	DNS_over_UDP	Edit Delete
3	☒ Allow	Any	Any	Any	Any	Edit Delete

☐ Allow Overrides

- e. Click **Save** to add the RADIUS server to the RADIUS server group.
3. Click **Save** again to save the RADIUS Server Group Object.

Run Remote Access VPN Wizard

In FMC, navigate to **Devices > VPN > Remote Access**. Click **Add**. This will launch the wizard.



Complete the Policy Assignment page of the wizard.

1. For Name enter **RAVPN**.
2. Uncheck **IPsec-IKEv2**
3. From Target Devices select **NGFW1**. Click **Add**.
4. Click **Next** in the bottom right corner.

The screenshot shows the 'Remote Access VPN Policy Wizard' at the 'Policy Assignment' step. The 'Name' field is 'RAVPN'. Under 'VPN Protocols', 'SSL' is checked and 'IPsec-IKEv2' is unchecked. Under 'Targeted Devices', 'NGFW1' is selected from the 'Available Devices' list. The 'Next' button at the bottom right is highlighted with a red box.

Choosing the authentication method for VPN Client

In the next page, the RA VPN wizard offers various methods of authentication - Primary Authentication using:

AAA only – Username/password authentication using RADIUS, LDAP or AD. In this lab ISE is used as RADIUS server to authenticate the VPN user credentials

Certificate Only – Each user is authenticated with a client certificate. The client certificate must be configured on VPN client endpoints. By default, the username is derived from the client certificate fields CN and OU. In this lab, the Anyconnect client machine already has device certificate installed and they are used to authenticate the identity of the VPN user. By default, the CN from the device certificate will be used to populate the username field of the authentication prompt in the Anyconnect Client

AAA + Certificate – A combination of ISE as RADIUS server and client certificate.

Note: The RA VPN configuration also offers Secondary authentication method. For example, you can set the primary authentication server to an LDAP or Active Directory realm and the secondary authentication to a RADIUS server. In this lab we will be using the Secondary Authentication method for MFA using DUO.

Complete the Connection Profile page of the wizard.

1. For Connection Profile Name enter **RAVPN**.
2. For the **Authentication Method** you can select **AAA Only**, **Client Certificate Only** or **Client Certificate & AAA**. For this lab exercise, we will use the method **Client Certificate & AAA**.
3. **Under Authentication, Authorization, Accounting (AAA)**, for Authentication and Authorization Server, select **ISE-RADIUS** (Authentication Server is available only if the **Authentication Method** is AAA only or AAA + Certificate).

Remote Access VPN Policy Wizard

1 Policy Assignment 2 **Connection Profile** 3 AnyConnect 4 Access & Certificate 5 Summary

Remote User AnyConnect Client Internet Outside VPN Inside Corporate Resources

Connection Profile:

Connection Profiles specify the tunnel group policies for a VPN connection. These policies pertain to creating the tunnel itself, how AAA is accomplished and how addresses are assigned. They also include user attributes, which are defined in group policies.

Connection Profile Name: **RAVPN**

This name is configured as a connection alias, it can be used to connect to the VPN gateway

Authentication, Authorization & Accounting (AAA):

Specify the method of authentication (AAA, certificates or both), and the AAA servers that will be used for VPN connections.

Authentication Method: **Client Certificate & AAA**

Username From Certificate: ☒ Map specific field ☐ Use entire DN (Distinguished Name) as username

Primary Field: **CN (Common Name)**

Secondary Field: **OU (Organisational Unit)**

Authentication Server: **ISE_RADIUS** (+)

Authorization Server: **ISE_RADIUS** (+)

Accounting Server: **ISE_RADIUS** (+)

Note: The Secondary Authentication steps are after this section.

4. Under **Client Address Assignment**, we create and assign the IPv4 Address Pool to be assigned to the Remote Access VPN users.

- a. Click on the **Pencil** icon. Click on the **+** icon. Enter the details as below:
 - i. For Name, enter **VPNPool**.
 - ii. For IPv4 Address Range, enter **198.19.10.57-198.19.10.62**.
 - iii. For Mask, enter **255.255.255.248**.
 - iv. Click **Save**

are defined in group policies.

Connection Profile Name: RAVPN

This name is configured as a connection alias, it can be used to connect to the VPN gateway.

Address Pools

Available IPv4 Pools: +

Selected IPv4 Pools:

Cancel OK

Add IPv4 Pool

Name: VPNPool

IPv4 Address Range: 198.19.10.57-198.19.10.62
Format: ipaddr-ipaddr e.g., 10.72.1.1-10.72.1.150

Mask: 255.255.255.248

Description:

☒ Allow Overrides

Configure device overrides in the address pool object to avoid IP address conflicts in case of object is shared across multiple devices

Override (0)

Cancel Save

Client Address Assignment:

Client IP address can be assigned from AAA server, DHCP server and IP address pools. When multiple options are selected, IP address assignment is tried in the order of AAA server, DHCP server and IP address pool.

☐ Use AAA Server (RADIUS only)

☐ Use DHCP Servers

☒ Use IP Address Pools

IPv4 Address Pools:

IPv6 Address Pools:

- b. From the Address Pools window select **VPNPool**, then click **Add** to add it to the **Selected IPv4 Pools** column. Click **OK**.

Address Pools

Available IPv4 Pools: +

Selected IPv4 Pools:

Cancel OK

5. Under the **Group Policy**, verify that the group policy is set to **DfltGrpPolicy**. Click on **Edit Group Policy**.

Client Address Assignment:

Client IP address can be assigned from AAA server, DHCP server and IP address pools. When multiple options are selected, IP address assignment is tried in the order of AAA server, DHCP server and IP address pool.

☐ Use AAA Server (RADIUS only) ●

☐ Use DHCP Servers

☒ Use IP Address Pools

IPv4 Address Pools: 

IPv6 Address Pools: 

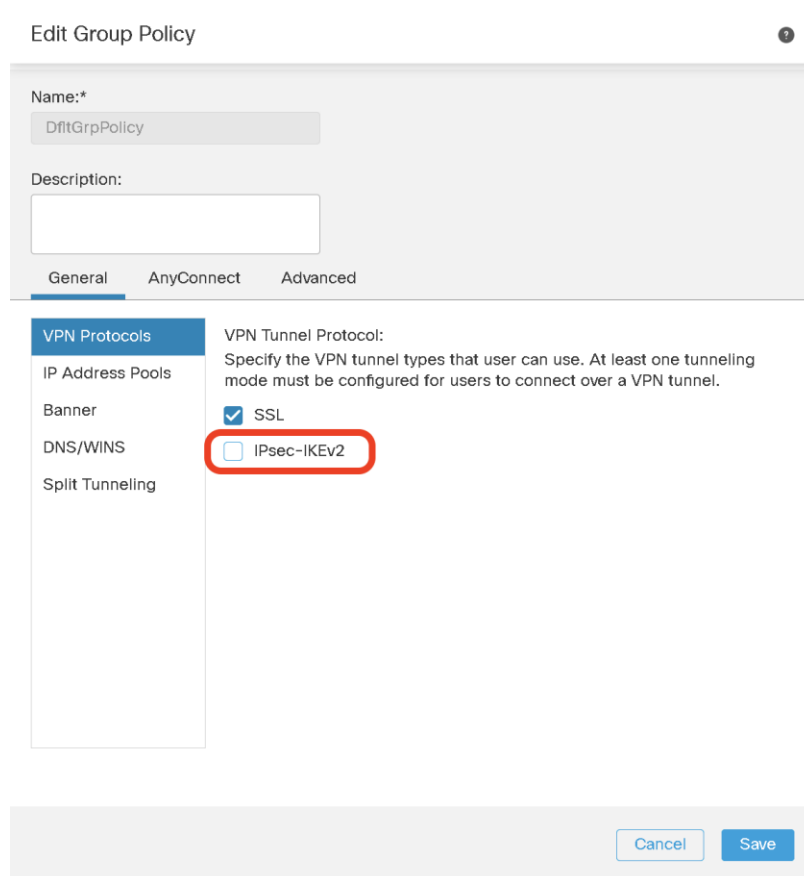
Group Policy:

A group policy is a collection of user-oriented session attributes which are assigned to client when a VPN connection is established. Select or create a Group Policy object.

Group Policy:* 
[Edit Group Policy](#)

NOTE: Typically, the VPN Group policy can be edited (or a new group policy is added) as a separate step. Here, we are making changes inline while running the RA VPN Wizard.

- a. Under **General > VPN Protocols** uncheck **IPsec-IKEv2**.



Edit Group Policy

Name:*
DfltGrpPolicy

Description:

General AnyConnect Advanced

VPN Protocols

VPN Tunnel Protocol:
Specify the VPN tunnel types that user can use. At least one tunneling mode must be configured for users to connect over a VPN tunnel.

☒ SSL

☐ IPsec-IKEv2

- b. Under **General > DNS/Wins**.
- i. Select **DNS_Server** from the Primary DNS Server drop-down list.

- ii. For Default Domain enter **dcloud.local**.

Edit Group Policy

Name:*
DfitGrpPolicy

Description:

General AnyConnect Advanced

VPN Protocols
IP Address Pools
Banner
DNS/WINS
Split Tunneling

Primary DNS Server:
DNS_Server +

Secondary DNS Server:
 +

Primary WINS Server:
 +

Secondary WINS Server:
 +

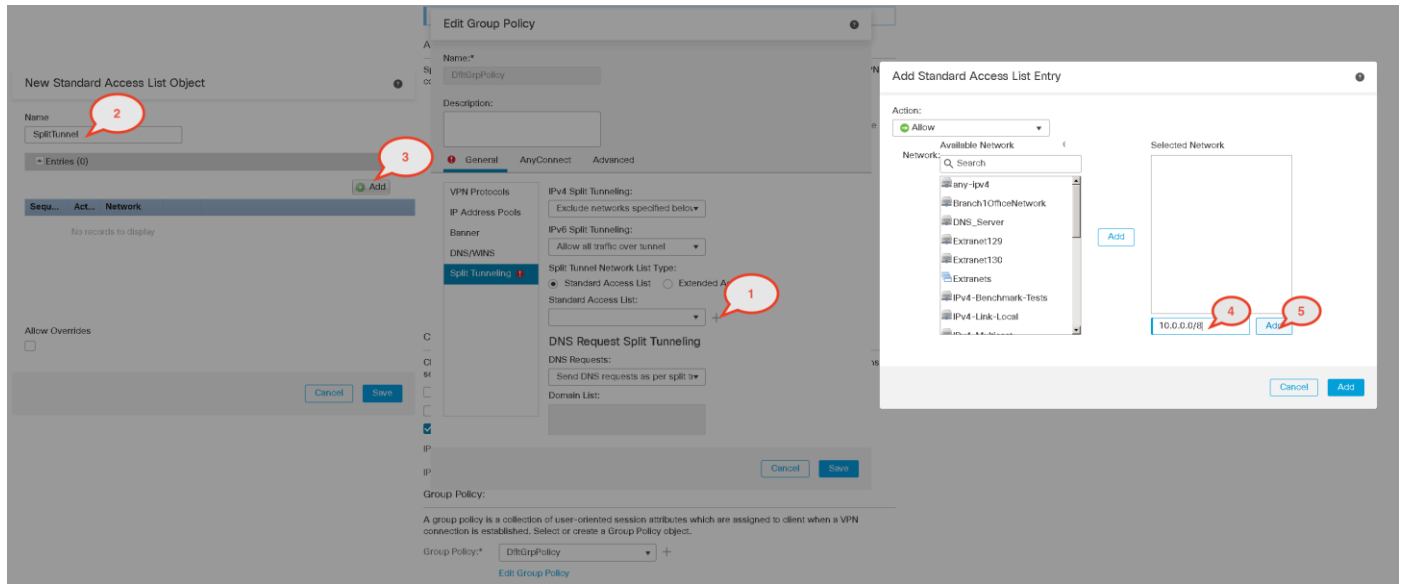
DHCP Network Scope:
 +
Only network object with ipv4 address is allowed (Ex: 10.72.3.5)

Default Domain:
dcloud.local

Cancel Save

NOTE: For best security, it is recommended that split-tunneling not be used. However, because there is no console access for the PC on which you will run AnyConnect, split tunneling must be used in this Scenario. Since there are different ways to access the PC in dCloud, you need to create a standard ACL to bypass all these potential access addresses. You will do this now.

- c. Under **General > Split Tunnel**.
- Select **Exclude networks specified below** from the IPv4 Split Tunneling drop-down list.
 - For the Standard Access List drop-down list, click **+**.
 - Create a standard access list called **SplitTunnel** with the ACE that allows **10.0.0.0/8**, **198.19.255.0/24** and **198.18.133.50**.
 - To do this, type the address in the textbox with the default text **Inline** and then Click **Add**.
 - Repeat** for all the addresses.
 - Click **Add** at the bottom right.
 - Click **Save** to save the access list.



New Standard Access List Object

Name: SplitTunnel

Entries (1)

Sequence No	Action	Network
1	Allow	10.0.0.0/8 198.19.255.0/24 198.18.133.50

☐ Allow Overrides

Cancel Save

Edit Group Policy

Name:*
DftGrpPolicy

Description:

General AnyConnect Advanced

VPN Protocols
IP Address Pools
Banner
DNS/WINS
Split Tunneling

IPv4 Split Tunneling:
Exclude networks specified below

IPv6 Split Tunneling:
Allow all traffic over tunnel

Split Tunnel Network List Type:
☒ Standard Access List
 ☐ Extended Access List

Standard Access List:
SplitTunnel

DNS Request Split Tunneling
 DNS Requests:
Send DNS requests as per split tr

Domain List:

Cancel Save

- d. Under **AnyConnect > Profiles**, click the **+** icon.
- i. Click **Browse** and select **AnyConnectProfile.xml** (the extension .xml may not be visible) from the **RA VPN** folder on the Jumpbox desktop. The remaining fields will auto-populate.

Add AnyConnect File

Name:*
AnyConnectProfile.xml

File Name:*
AnyConnectProfile.xml **Browse..**

File Type:*
AnyConnect Client Profile

Description:

Cancel Save

- ii. Click **Browse** and select the **anyconnect-win-4.7.02036-webdeploy-k9.pkg** from the **RA VPN** folder on the Jumpbox desktop. The remaining fields will auto-populate.

Add AnyConnect File

Name:*

File Name:*
 [Browse..](#)

File Type:*

Description:

[Cancel](#) [Save](#)

- e. select **AnyConnectProfile.xml** from the Client Profile drop-down list.

Edit Group Policy

Name:*

Description:

General **AnyConnect** Advanced

Profiles

SSL Settings

Connection Settings

AnyConnect profiles contains settings for the VPN client functionality and optional features. FTD deploys the profiles during AnyConnect client connection.

Client Profile:
 +

Standalone profile editor can be used to create a new or modify existing AnyConnect profile. You can download the profile editor from [Cisco Software Download Center](#).

[Cancel](#) [Save](#)

6. Click **Save** to save the changes you made to DfltGrpPolicy.
7. Click **Next**

Complete the AnyConnect page of the wizard.

Check the file object checkbox and click **Next**.

Remote Access VPN Policy Wizard

1 Policy Assignment — 2 Connection Profile — **3 AnyConnect** — 4 Access & Certificate — 5 Summary

AnyConnect Client Image

The VPN gateway can automatically download the latest AnyConnect package to the client device when the VPN connection is initiated. Minimize connection setup time by choosing the appropriate OS for the selected package.

Download AnyConnect Client packages from [Cisco Software Download Center](#).

Show Re-order buttons +

☐	AnyConnect File Object Name	AnyConnect Client Package Name	Operating System
☒	anyconnect-win-4.7.02036-we...	anyconnect-win-4.7.02036-webdeploy-k9...	Windows

Complete the Access & Certificate page of the wizard

In this page of the wizard, we choose the NGFW interface which will accept Incoming VPN connections. Also, we select the Server-Side certificate the NGFW will present for VPN connection.

1. For Interface group/Security Zone select **OutZone**.
2. For Certificate Enrollment, click +.
 - a. For Name enter **NGFW1_Outside**.
 - b. Select **PKCS12 File** from the Enrollment Type drop-down menu.
 - c. Click **Browse** and select **ngfw-dcloud.pfx** (the extension may be hidden) from the **Certificates\Lab Certificates\Other Certificates** folder on the Jumpbox desktop.
 - d. For Passphrase enter **C1sco12345**.
 - e. Click **Save**

Add Cert Enrollment

Name*

Description

CA Information Certificate Parameters Key Revocation

Enrollment Type:

PKCS12 File*:
 [Browse PKCS12 File](#)

Passphrase:

☐ Allow Overrides

[Cancel](#) [Save](#)

3. Select **NGFW1_Outside** from the dropdown.

4. Click **Next**.

Remote Access VPN Policy Wizard

1 Policy Assignment 2 Connection Profile 3 AnyConnect 4 Access & Certificate 5 Summary

Remote User → AnyConnect Client → Internet → Outside → VPN Device → Inside → Corporate Resources

AAA

Network Interface for Incoming VPN Access
 Select or create an Interface Group or a Security Zone that contains the network interfaces users will access for VPN connections.
 +
☒ Exclude DUTs on provider interfaces
 All the devices must have interfaces as part of the Interface Group/Security Zone selected.

Device Certificates
 Device certificate (also called Identity certificate) identifies the VPN gateway to the remote access clients. Select a certificate which is used to authenticate the VPN gateway.
 +
☒ Enroll the selected certificate option on the target devices

Access Control for VPN Traffic
 All decrypted traffic in the VPN tunnel is subjected to the Access Control Policy by default. Select this option to bypass decrypted traffic from the Access Control Policy.
☐ Bypass Access Control policy for decrypted traffic (sysopst permit-vpn)
 This option bypasses the Access Control Policy inspection, but VPN filter ACL and authorization ACL, downloaded from AAA server are still applied to VPN traffic.

Note: We are not selecting the option 'Bypass Access Control Policy for decrypted traffic'. If you select this option, we don't need to create specific rules in the Access Control Policy to allow the decrypted VPN traffic. For the purpose of this lab, we will be modifying the Access Control to allow the VPN traffic after decryption.

Review the Configuration on the **Summary** page and click **Finish**.

Click Finish.

Remote Access VPN Policy Wizard

1 Policy Assignment 2 Connection Profile 3 AnyConnect 4 Access & Certificate 5 **Summary**

Remote Access VPN Policy Configuration

Firepower Management Center will configure an RA VPN Policy with the following settings

Name: RAVPN
 Device Targets: NGFW1
 Connection Profile: RAVPN
 Connection Alias: RAVPN
 AAA:
 Authentication Method: AAA Only
 Authentication Server: ISE_RADIUS
 Authorization Server: ISE_RADIUS
 Accounting Server: -
 Address Assignment:
 Address from AAA: -
 DHCP Servers: -
 Address Pools (IPv4): VPNPool
 Address Pools (IPv6): -
 Group Policy: DfGrpPolicy
 AnyConnect Images: anyconnect-win-4.7.02036-webdeploy-k9.pkg
 Interface Objects: OutZone
 Device Certificates: NGFW1_Outside

Device Identity Certificate Enrollment

Certificate enrollment object 'NGFW1_Outside' is not installed on one or more targeted devices. Certificate installation will be initiated on the targeted devices on finishing the wizard. Go to the [Certificates](#) page to check the status of the installation.

Additional Configuration Requirements

After the wizard completes, the following configuration needs to be completed for VPN to work on all device targets.

- Access Control Policy Update**
An **Access Control** rule must be defined to allow VPN traffic on all targeted devices.
- NAT Exemption**
If NAT is enabled on the targeted devices, you must define a **NAT Policy** to exempt VPN traffic.
- DNS Configuration**
To resolve hostname specified in AAA Servers or CA Servers, configure DNS using [FlexConfig Policy](#) on the targeted devices.
- Port Configuration**
SSL will be enabled on port 443. Please ensure that these ports are not used in **NAT Policy** or other services before deploying the configuration.
- Network Interface Configuration**
Make sure to add interface from targeted devices to SecurityZone object 'OutZone'

Cancel Back **Finish**

Adding Secondary Authentication for Remote Access VPN Users using DUO MFA (Optional)

This section is optional. If you don't wish to add a Secondary authentication mechanism, please proceed with the steps in the next section titled **Allow VPN Traffic through the NGFW**.

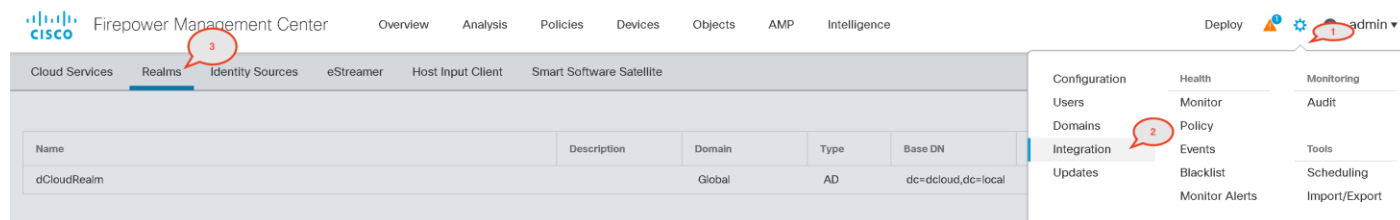
- This section showcases two methods of DUO integrations. You can do either of the next two sections depending on the type of Duo Integration you want to implement.
 - Section A - Create and configure and LDAP server to connect to Duo. This method uses a direct LDAPS connection between the firewall and DUO Cloud. For the LDAPS method, you will need to keep the Integration Key, Secret Key and API Hostname from the Duo Application Dashboard handy.
 - Section B - Create and configure a RADIUS Server object for Duo Authentication Proxy. This method makes use of the Authentication Proxy server that was already set up for the purposes of the lab.
- Complete the DUO account setup steps in Appendix A. This is done irrespective of the type of DUO integration being used.

3. Proceed to either of the next Section based on the choice of DUO integration.

Note: Skip this section and jump to Section B, if you wish to use DUO Authentication Proxy for RADIUS Authentication

Section A - Create and configure an LDAP server to connect to Duo

1. Navigate to **System > Integration > Realms**. Click on **New Realm**.
 - a. Call the realm **Duo-LDAPS**
 - b. Change Type to **LDAP**
 - c. Directory Username will be of the format `dc=INTEGRATION_KEY,dc=duosecurity,dc=com`
 - d. Directory Password will be the Secret Key.
 - e. Base DN and Group DN will also be of the format `dc=INTEGRATION_KEY,dc=duosecurity,dc=com`
 - f. Leave other attributes to default and click on OK.



Add New Realm

Name *

Description

Type *

LDAP ▼

Directory Username *

ex:

uid=user,dc=example,dc=com

Directory Password *

Base DN *

ex:

ou=user,dc=cisco,dc=com

Group DN *

ex:

ou=group,dc=cisco,dc=com

Group Attribute

Unique Member ▼

Cancel

OK

Note: The above screenshot only shows dummy values as they will be different for every lab user

2. Click on Add directory.
 - a. Hostname/IP address will be your API Hostname
 - b. Select **LDAPS** as Encryption. Select **DigiCert-High-Assurance** as the SSL Certificate.
 - c. Click on **Test**. This should succeed and click on OK Click **save**
 - d. If the test is unsuccessful, please verify the following settings:
 - i. Recheck the LDAP realm settings with the DUO Admin page and recheck the DUO API value in the hostname.
 - ii. Make sure that the DNS settings for the FMC are correct. They should be present under **System > Configuration > Management Interfaces**

Add directory

Hostname / IP Address
api-9999999.duosecurity.com

Port
636

Encryption
☐ STARTTLS
☒ LDAPS
☐ None

SSL Certificate
DigiCert-High-Assurance-EV-Rc +

Cancel Test OK

Add directory

Hostname / IP Address
api-38238671.duosecurity.com

Port
636

Encryption
☐ STARTTLS
☒ LDAPS
☐ None

SSL Certificate
DigiCert-High-Assurance-EV-Rc +

Cancel Test OK

Status

Test connection succeeded

OK

Note: The above screenshot only shows dummy values as they will be different for every lab user.

3. Click on **Save**. Then enable the state of the realm.

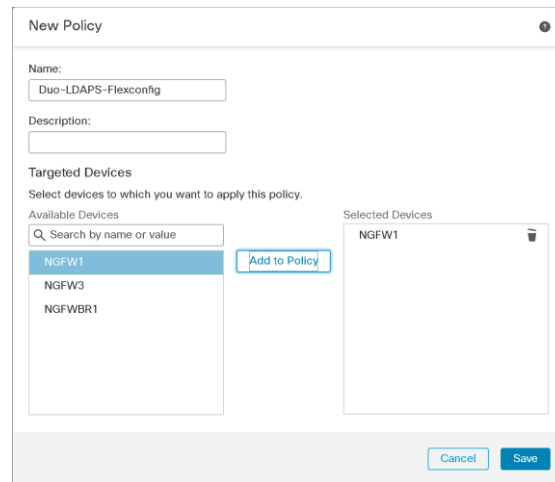
uniqueMember



4. Duo requires CN to be the LDAP Naming Attribute. Since we could not configure this on the LDAP Server UI, we will make use of Flexconfig. Navigate to **Devices > FlexConfig**.

5. Click on Add a new Policy.

- a. Set the name to Duo-LDAPS-Flexconfig.
- b. Add **NGFW1** as a device and click **Save**.



New Policy

Name: Duo-LDAPS-Flexconfig

Description:

Targeted Devices
Select devices to which you want to apply this policy.

Available Devices: NGFW1, NGFW3, NGFWBR1

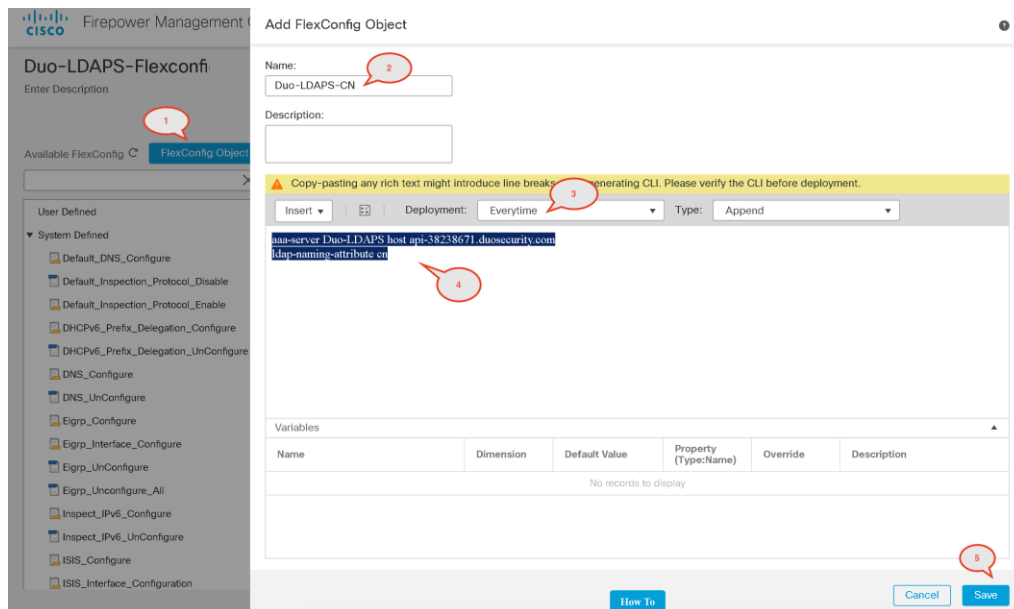
Selected Devices: NGFW1

Buttons: Cancel, Save

6. Click on button **FlexConfig Object**.
 - a. Set Name to **Duo-LDAPS-CN**
 - b. Change Deployment to **Everytime**
 - c. Add the following configuration lines:

```
aaa-server Duo-LDAPS host <API hostname>
  ldap-naming-attribute cn
```

- d. Click on **Save**.



Add FlexConfig Object

Name: Duo-LDAPS-CN

Description:

Available FlexConfig: FlexConfig Object

User Defined: System Defined

Configuration Lines:

```
aaa-server Duo-LDAPS host api-38238671.duosecurity.com
  ldap-naming-attribute cn
```

Deployment: Everytime Type: Append

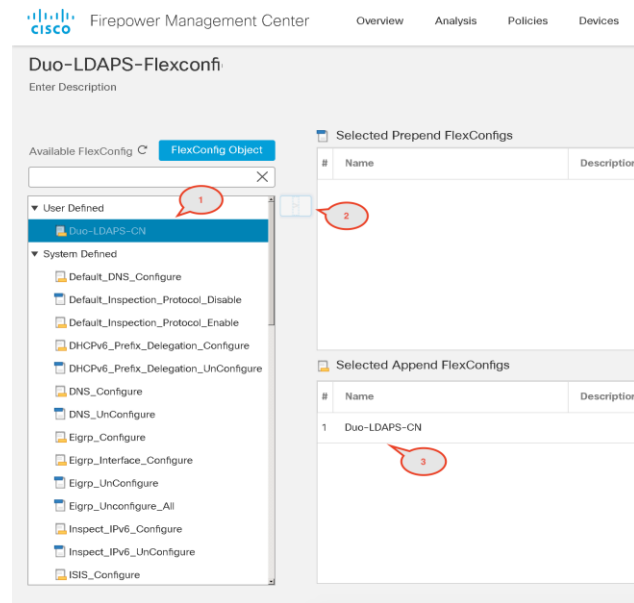
Variables:

Name	Dimension	Default Value	Property (Type:Name)	Override	Description
No records to display					

Buttons: How To, Cancel, Save

Note: It is important to select Deployment as **Everytime** as the FMC might treat this configuration as OOB changes and remove this configuration upon next deployment.

7. Select this newly created object from the **Available FlexConfig** in the left-hand navigation pane. Click on **>**. This will add the FlexConfig object to the appended section. Click on **Save**.



Note: Here the FlexConfig object is added to the **Append** (default) section. The commands defined by the FlexConfig object will be appended to the configuration generated by the Firepower Management Center upon deployment. The **Prepend** section is normally used for putting FlexConfig object at the beginning of the configurations generated from the Firepower Management Center policies. E.g. for commands that clear or negate a configuration.

Section B - Create and configure a RADIUS Server object for Duo Authentication Proxy

1. For convenience, we have already configured the Duo Authentication Proxy application on the internal Active Directory Server. You can open an RDP session to this server to take a look at the Auth Proxy configuration file. This file is located at **C:\Program Files (x86)\Duo Security Authentication Proxy\conf**.

```

; Complete documentation about the Duo Auth Proxy can be found
here:
; https://duo.com/docs/authproxy_reference

; MAIN: Include this section to specify global configuration
options.
; Reference: https://duo.com/docs/authproxy_reference#main-
section
;[main]

; CLIENTS: Include one or more of the following configuration
sections.
; To configure more than one client configuration of the same
type, append a
; number to the section name (e.g. [ad_client2])

[duo_only_client]

; SERVERS: Include one or more of the following configuration
sections.
; To configure more than one server configuration of the same
type, append a
; number to the section name (e.g. radius_server_auto1,
radius_server_auto2)

[radius_server_auto]
ikey=DIB86UGGW1KH44Y7LJCH
skey=aHGHby4M4f8crnmwDuXzQ80M0ADtHxgtRJcmzq7lO
api_host=api-29acc416.duosecurity.com
radius_ip_1=198.19.10.1
radius_secret_1=C1sco12345
radius_ip_2=198.19.10.2
radius_secret_2=C1sco12345
radius_ip_3=198.19.10.3
radius_secret_3=C1sco12345
failmode=safe
client=duo_only_client

```

2. Update the configuration file with the values from the DUO Account noted earlier for the Application – **Firepower Threat Defense VPN**
 - a. ikey = Integration Key
 - b. skey = Secret Key
 - c. api = API hostname
 - d. Save and close the file once done.
3. On the AD Server, open up **Services.msc** and restart services for **Duo Authentication Proxy**. Make sure that this is successful.
4. Navigate to C:\Program Files (x86)\Duo Security Authentication Proxy\bin and run authproxy_connectivity_tool.
5. To see if the connection tests were successful, navigate to C:\Program Files (x86)\Duo Security Authentication Proxy\log and open the connectivity_tool file. Scroll to the bottom of the file and confirm that you see the following logs:

```

2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] Testing section 'duo_only_client' with
configuration:
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] {}
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] There are no configuration problems

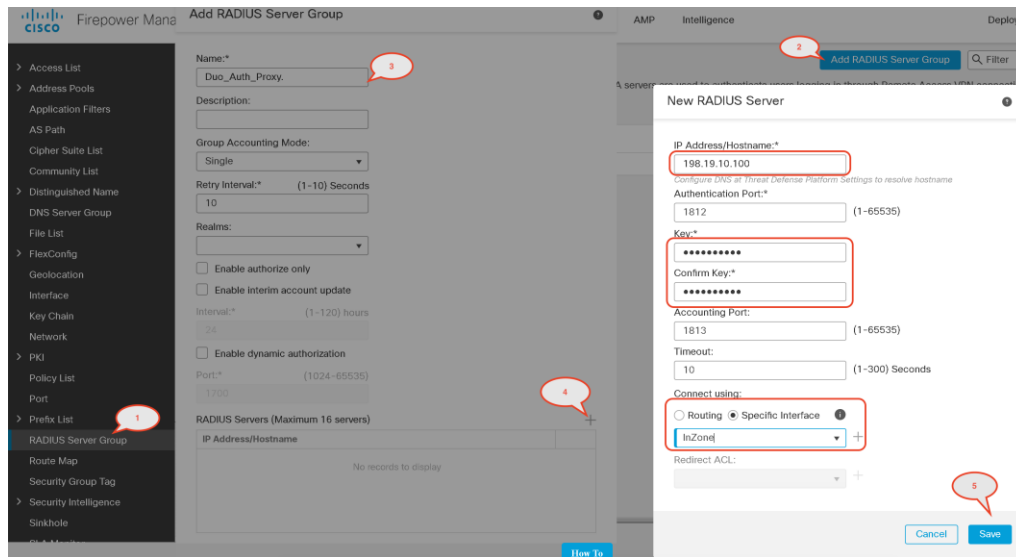
```

```

2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] Testing section 'radius_server_auto' with
configuration:
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] {'api_host': 'api-38238671.duosecurity.com',
'client': 'duo_only_client',
'failmode': 'safe',
'key': 'DI4RHEKLY5WQ0JO8EZ8U',
'port': '1812',
'radius_ip_1': '198.19.10.1',
'radius_ip_2': '198.19.10.2',
'radius_ip_3': '198.19.10.3',
'radius_secret_1': '*****',
'radius_secret_2': '*****',
'radius_secret_3': '*****',
'skey': '*****[40]'}
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] There are no configuration problems
2020-01-14T17:15:16+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] Testing section 'duo_only_client' with
configuration:
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] {}
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] No testing to be done for section.
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] Testing section 'radius_server_auto' with
configuration:
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] {'api_host': 'api-38238671.duosecurity.com',
'client': 'duo_only_client',
'failmode': 'safe',
'key': 'DI4RHEKLY5WQ0JO8EZ8U',
'port': '1812',
'radius_ip_1': '198.19.10.1',
'radius_ip_2': '198.19.10.2',
'radius_ip_3': '198.19.10.3',
'radius_secret_1': '*****',
'radius_secret_2': '*****',
'radius_secret_3': '*****',
'skey': '*****[40]'}
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] The RADIUS Server has no connectivity problems.
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] -----
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] SUMMARY
2020-01-14T17:15:21+0000 [duoauthproxy.lib.log#info] No issues detected

```

6. On the FMC, navigate to **Objects > RADIUS Server Group**.
7. Click Add RADIUS Server Group.
 - a. Call the group **Duo_Auth_Proxy**.
 - b. Click the **+** sign to add a RADIUS server at the bottom of the window.
8. Enter the following information, leaving other attributes as default.
 - a. For IP Address/Hostname, enter **198.19.10.100**.
 - b. For Key, enter **C1sco12345**.
 - c. Select the Specific interface radio button and select **InZone1** from the drop-down menu.



9. Click **Save** to add the RADIUS server to the RADIUS server group. When you are done, your RADIUS server group should look like the following. Click **Save**.

RADIUS Servers (Maximum 16 servers)	
IP Address/Hostname	
198.19.10.100	 

Duo configuration for the RAVPN profile

We will be adding DUO now as the secondary authentication server.

1. Go to **Devices > VPN > Remote Access > RAVPN**
2. Under the Connection Profile tab, edit the **RAVPN** profile.
 - a. Navigate to the **AAA** tab and select **Use secondary authentication**.
 - b. Expand Map username from client certificate. Select Prefill username from certificate on user login window.
 - c. Depending on your chosen method of Duo Integration, select either **Duo-LDAPS** or **Duo_Auth_Proxy** as your Authentication Server. The following steps use the **Duo-LDAPS** server.
 - d. Expand Username for secondary authentication and select Map username from client certificate.
 - e. Select Prefill username from certificate on user login window. Select Hide username in login window.
 - f. Click on **Save**

3. Click on **Save** to save pending changes.

Allowing VPN Traffic through the NGFW

Modify the Access control Policy to allow decrypted VPN traffic

1. In FMC, navigate to **Policies > Access Control > Access Control**.
2. Select and edit the access control policy **Base_Policy**. Click **Add Rule**.
 - a. For Name enter **AnyConnect-S3-Permit**.
 - b. Select **into Default** from the Insert drop-down list.
 - c. Keep the action as **Allow**.
 - d. The Zones tab should already be selected.
 - e. Select **OutZone** and click **Add to Source**.
 - f. Select **InZone1** and click **Add to Destination**.

Note: The direction of the flow of the VPN traffic is from OutZone to InZone1 as the VPN is terminating on the Outside interface which is assigned zone **OutZone**

3. Select the **Networks** tab.
 - a. Search and select **VPNPoolIPs** and click **Add to Source**.
 - b. Search and select **LAN_Network** and click **Add to Destination**.
4. Select the **Inspection** tab.

- | | | | | | | | | | | | | | | | | | | | | | | |
|----|------------------|---------|--------|------------|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-------|--|--|--|--|--|---|--|--|
| 10 | AnyConnect-S3... | OutZone | InZone | VPNPoolIPs | LAN_Network | Any | Any | Any | Any | Any | Any | Any | Any | Allow | | | | | | 0 | | |
|----|------------------|---------|--------|------------|-------------|-----|-----|-----|-----|-----|-----|-----|-----|-------|--|--|--|--|--|---|--|--|

- NOTE: Enabling Do not proxy ARP on Destination Interface is critical in this lab exercise. If you miss this step, your pod may have access issues, since all devices are managed in band.

- NOTE: The last change was done to allow for Posture Assessment to trigger. As Posture probes from Anyconnect Client will be internet bound to enroll.cisco.com, the traffic has to hit the NGFW outside interface and do a u-turn. This modification will ensure that the probes are translated to the outside interface IP for internet routing.

					Original Packet			Translated Packet				
#	Direction	Type	Source Interface	Destination Interface	Original Sources	Original Destinations	Original Services	Translated Sources	Translated Destinations	Translated Services	Options	
NAT Rules Before												
1		Static	InZone	OutZone	LAN_Network	VPNPoolIPs		LAN_Network	VPNPoolIPs		Dns:false no-proxy-arp	
Auto NAT Rules												
#		Static	InZone	OutZone	wwwin			wwwout			Dns:false	
NAT Rules After												
2		Dyna...	any	OutZone	any			Interface			Dns:false	

Configure DNS Settings for NGFW1

This will be used for DNS resolution by the device in order to reach ISE FQDN or DUO FQDN (in a different scenario).

1. Select **DNS Server Group**. Click on **Add DNS Server Group**.
 - a. Enter Name as **DCloud-DNS**.
 - b. Default Domain will be **dcloud.local**
 - c. Enter **198.19.10.100** as the DNS Server.
 - d. Click on **Save**.

New DNS Server Group Object

Name*:

Default Domain:

Timeout:

Range: 1 - 30 Seconds

Retries:

Range: 0 - 10

DNS Servers:

(Multiple values in IPv4 or IPv6 addresses can be specified as comma separated entries)

Modify the Platform Settings

The platform setting policy is being modified to add DNS server group we created earlier to the FTD.

1. Navigate to **Devices > Platform Settings** and edit the **NGFW1_Platform_Settings**. This platform setting policy should be assigned to **NGFW1**.

Policy Assignments ?

Targeted Devices

Select devices to which you want to apply this policy.

Available Devices

NGFW1

NGFW3

NGFWBR1

Add to Policy

Selected Devices

NGFW1

Cancel OK

2. In the Platform Settings window, navigate to DNS.
 - a. Check **Enable DNS name resolution by device**.
 - b. Select **DCloud-DNS** from the DNS Server Group dropdown.
 - c. Add **InZone1** as the Interface Object.
 - d. Click on **Save**.

- ARP Inspection
- Banner
- DNS**
- External Authentication
- Fragment Settings
- HTTP
- ICMP
- Secure Shell
- SMTP Server
- SNMP
- SSL
- Syslog
- Timeouts
- Time Synchronization
- UCAPL/OC Compliance

DNS Resolution Settings

Specify DNS servers group and device interfaces to reach them.

☒ Enable DNS name resolution by device

DNS Server Group: DCloud-DNS +

Expiry Entry Timer: 1 Range: 1-65535 minutes

Poll Timer: 240 Range: 1-65535 minutes

Interface Objects

Devices will use specified interface objects for connecting with DNS Servers.

Available Interface Objects

InZone

OutZone

Add

Selected Interface Objects

InZone

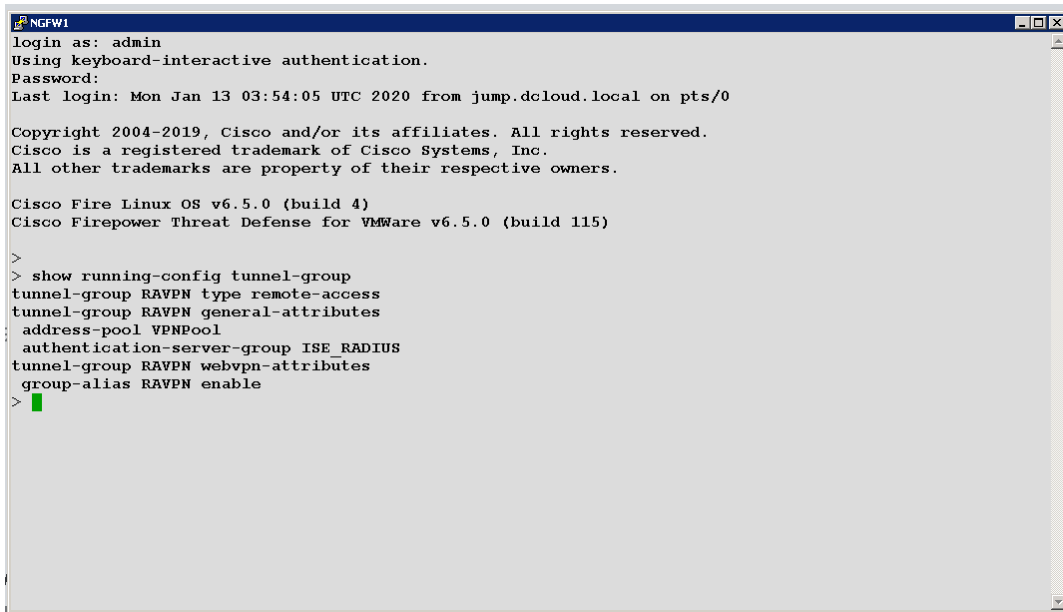
☐ Enable DNS Lookup via diagnostic interface also.

Deploy and Test VPN Connectivity

Deploy the changes from the FMC

1. Deploy policy to device. In FMC, click the **Deploy** button.

2. Select **NGFW1** and Click **Deploy**.
3. Wait for the deployment to complete. You can monitor the status of the deployment by clicking on the **Notifications** tab.
4. After the deployment is complete, from the Jumpbox open a PUTTY session to the **NGFW1 CLI**. Log in as **admin/C1sco12345** and run some or all of the following commands.
 - a. `show running-config tunnel-group`



```

NGFW1
login as: admin
Using keyboard-interactive authentication.
Password:
Last login: Mon Jan 13 03:54:05 UTC 2020 from jump.dcloud.local on pts/0

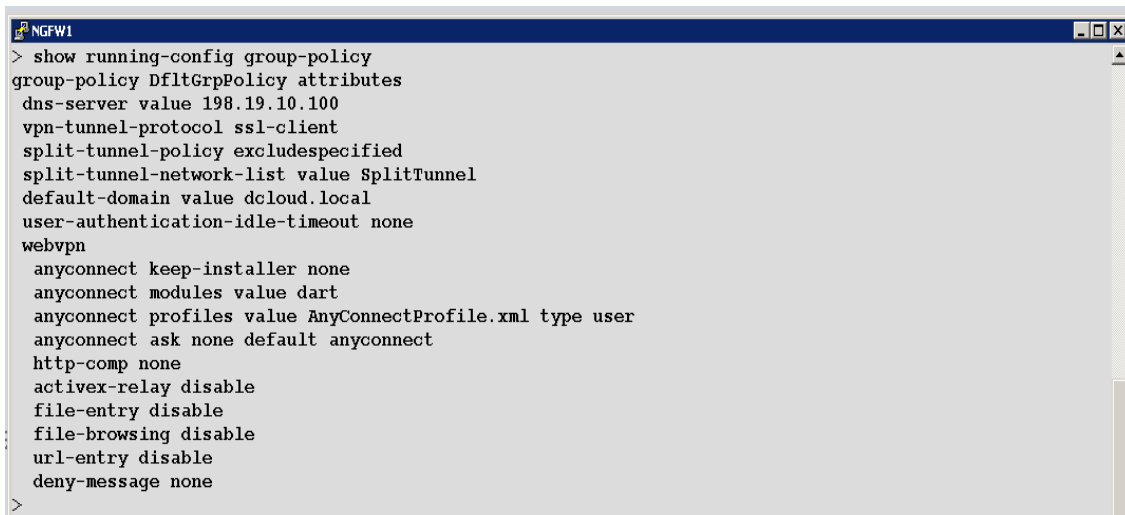
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All other trademarks are property of their respective owners.

Cisco Fire Linux OS v6.5.0 (build 4)
Cisco Firepower Threat Defense for VMWare v6.5.0 (build 115)

>
> show running-config tunnel-group
tunnel-group RAVPN type remote-access
tunnel-group RAVPN general-attributes
address-pool VPNPool
authentication-server-group ISE_RADIUS
tunnel-group RAVPN webvpn-attributes
group-alias RAVPN enable
>

```

- b. `show running-config group-policy`



```

NGFW1
> show running-config group-policy
group-policy DfltGrpPolicy attributes
dns-server value 198.19.10.100
vpn-tunnel-protocol ssl-client
split-tunnel-policy excludespecified
split-tunnel-network-list value SplitTunnel
default-domain value dcloud.local
user-authentication-idle-timeout none
webvpn
anyconnect keep-installer none
anyconnect modules value dart
anyconnect profiles value AnyConnectProfile.xml type user
anyconnect ask none default anyconnect
http-comp none
activex-relay disable
file-entry disable
file-browsing disable
url-entry disable
deny-message none
>

```

- c. `show running-config crypto`. Notice the crypto trust point name is: **NGFW1_Outside**

```
> show running-config crypto
crypto ipsec security-association pmtu-aging infinite
crypto ca trustpoint NGFW1_Outside
  keypair NGFW1_Outside
  crl configure
crypto ca trustpool policy
crypto ca certificate chain NGFW1_Outside
```

d. show running-config ip local pool

```
> show running-config ip local pool
ip local pool VPNPool 198.19.10.57-198.19.10.62 mask 255.255.255.248
```

e. show running-config nat

```
> show running-config nat
nat (inside,outside) source static LAN Network LAN Network destination static VPNPoolIPs VPNPoolIPs no-proxy-arp
!
object network wwwin
  nat (inside,outside) static wwwout
!
nat (any,outside) after-auto source dynamic any interface
```

5. Test AAA by running the following command on the NGFW1 CLI:

test aaa-server authentication ISE_RADIUS host 198.19.10.130 username vpnuser password C1sco12345

```
> test aaa-server authentication ISE_RADIUS host 198.19.10.130 username vpnuser password C1sco12345
INFO: Attempting Authentication test to IP address (198.19.10.130) (timeout: 12 seconds)
INFO: Authentication Successful
```

The outcome of the above command should be **Authentication Successful**. If the output is not successful, please verify the configuration in the section named **Create and configure a RADIUS server group for ISE**.

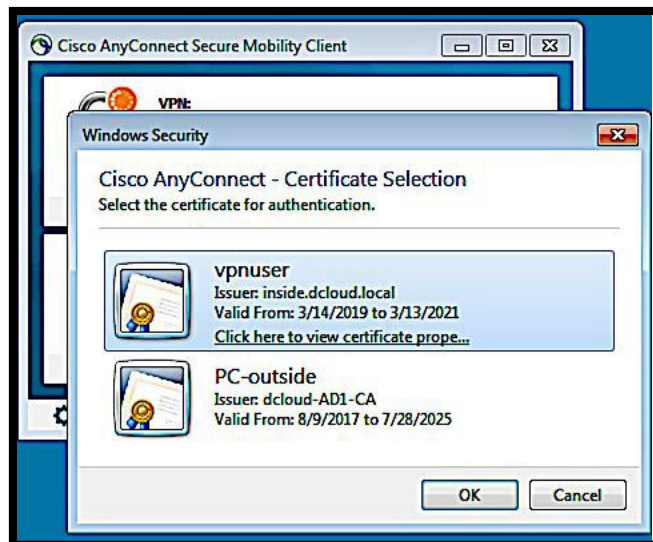
Connect with AnyConnect

NOTE: In this scenario, the definition of a compliant system is a system that has a file called `compliant.txt` on the desktop. In this exercise, `Wkst2` will start out as non-compliant. And furthermore, the posture module is already installed on `Wkst2`.

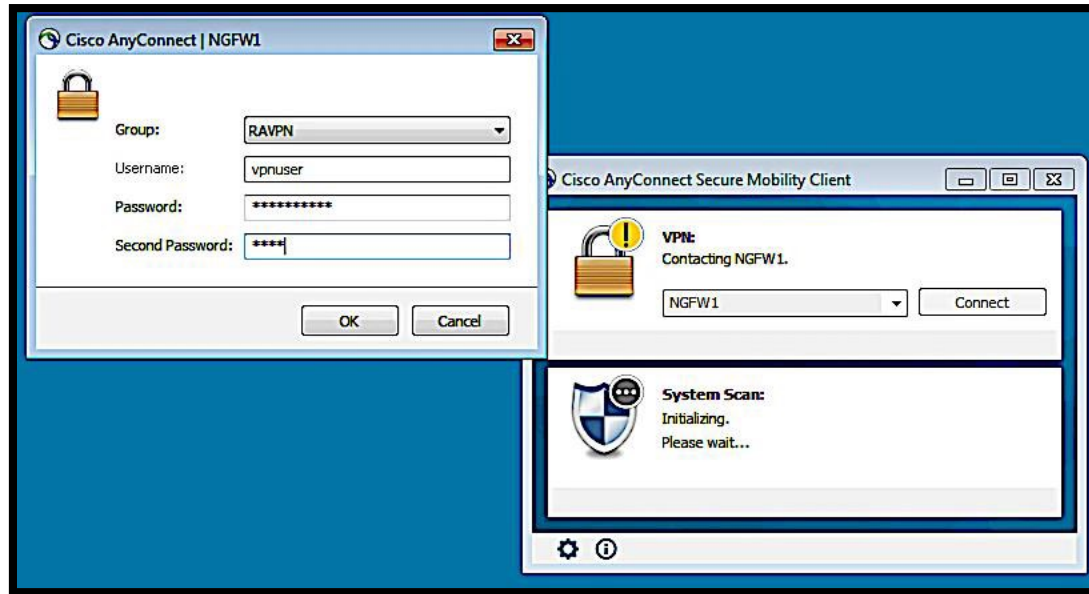
1. Connect to **Wkst2**. You will automatically be logged in as the administrator. You can connect using one of two methods.
 - a. From the topology map, as show in this picture. This is the recommended method.



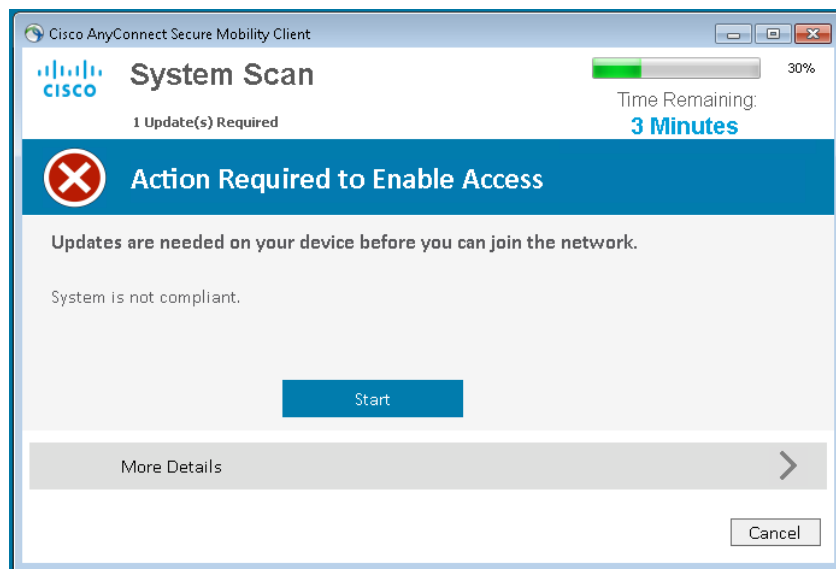
- b. By clicking the **Wkst2** (Outside PC) shortcut in the Remote Desktops folder the Jumpbox desktop. However, if you do this, you must allow local LAN access in the AnyConnect client.
2. If you are connected to the pod via VPN, use an RDP client on your laptop to connect to 198.18.133.23. Log in as Administrator, password C1sco12345.
3. From the desktop of **Wkst2** open **AnyConnect** from the Start Menu. The **Connect To** field should have **NGFW1** selected. Click on **Connect**. When prompted log in as **vpnuser** with password **C1sco12345**. For testing we are going to use **vpnuser** and a client certificate of the same name. Select the vpnuser certificate

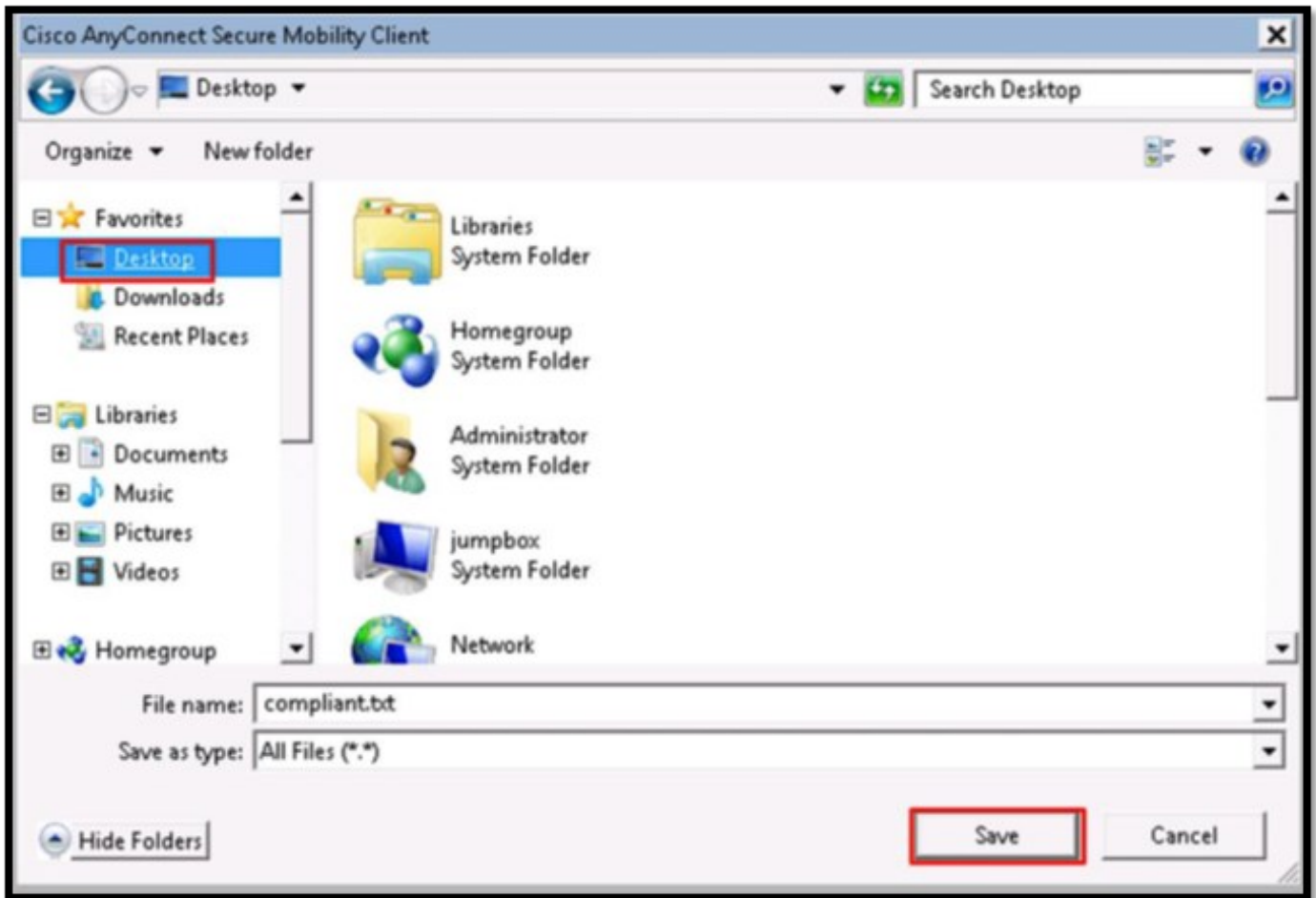


4. Enter the password as **C1sco12345** and **second password as push**. Click on OK. Push is because we are going to use the Push method of Duo authentication, but this can be set to phone or SMS if you want. You should get a push notification on your mobile device to approve and when you do this, you will see the VPN tunnel getting established.

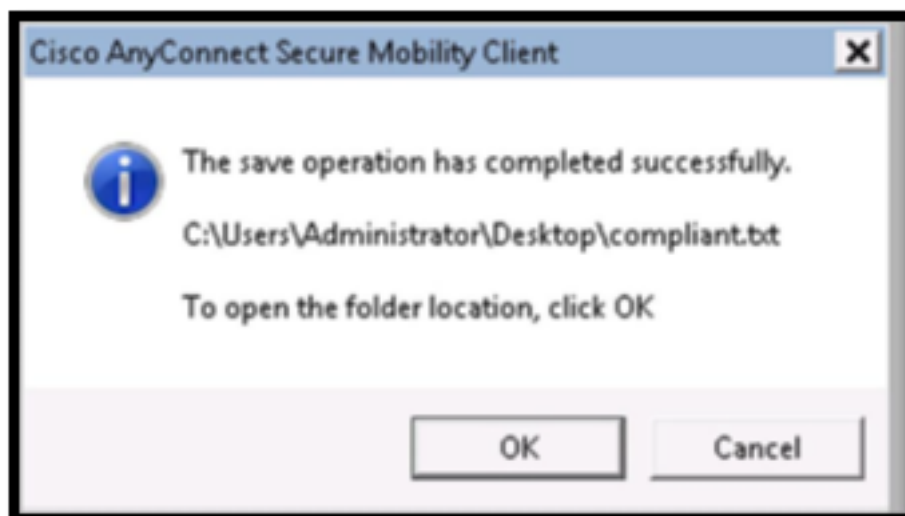


5. On the first connect, you will see the AnyConnect Compliance module being downloaded.
 - a. Since the system will not be compliant the first time we connect, you will be prompted to remediate. Click on **Start**.
 - b. The module will prompt you save the **compliant.txt** file. The window to save the **compliant.txt** file might appear behind the System Scan window. Change the destination folder to the **Desktop**.

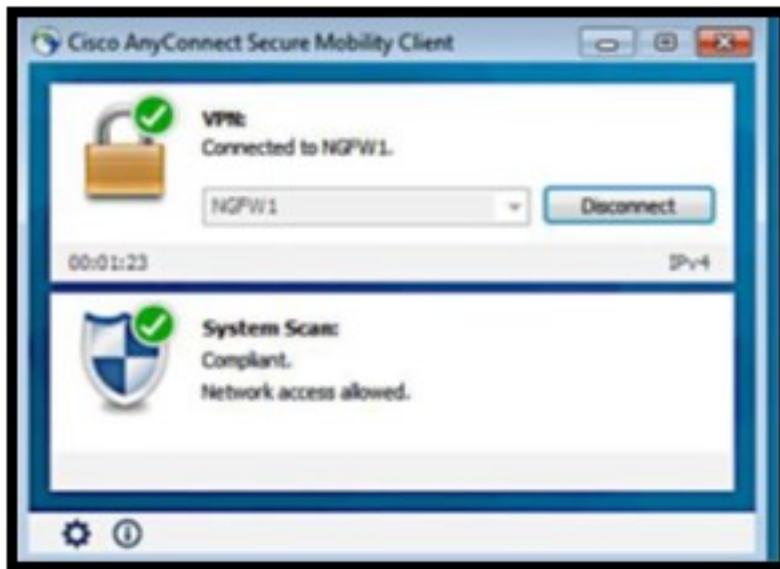




6. In the dialog box that says **The save operation has complete successfully**, you can click **Cancel**. There is no need to open the folder.



7. After the file is installed, the client will now declare your system compliant which you can confirm by looking in the system tray and clicking on the Cisco AnyConnect icon (a globe with a padlock). The following will open.



8. Using the bookmarks in the Firefox browser on Wkst2, confirm that you can now access two internal websites that are bookmarked: **Inside**, **Alt Inside**.
9. In any one of these internal servers click on the **Files** link, then click on **Zombies.pdf**, which is considered malware. Confirm that it is blocked. Confirm that a benign file like **ProjectX.pdf**, is not blocked.

NOTE: To make sure that a cloud lookup time out (which happens occasionally in these pods) does not break the exercise, the file **Zombies.pdf** was added to the FMC custom detection list. If you want to perform a test that actually requires the cloud lookup, enter the URL <http://altoutside.dcloud.local/malware>, and try to download **Buddy.exe**.

10. Confirm that intrusions are being blocked. This can be done by opening a command prompt on **Wkst2** and masking an FTP connection to **inside.dcloud.local**. The connection should be allowed. Log in as **guest**, password **C1sco12345**. Once logged in type **cd ~root**. The connection should be reset because Snort signature 336 has been triggered.


```

Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ftp inside.dcloud.local
Connected to inside.dcloud.local.
220 (vsFTPd 3.0.2)
User (inside.dcloud.local:(none)): guest
331 Please specify the password.
Password:
230 Login successful.
ftp> cd ~root
Connection closed by remote host.
ftp> _

```

11. (Optional) In the FMC window opened on the jumpbox, examine the malware event (**Analyze > Files > Malware Events**) and intrusion event (**Analyze > Intrusions > Events**). You can also examine RA VPN user statistics by navigating to **Overview > Dashboards > Access Controlled User Statistics > VPN**.
12. On **NGFW1** PUTTY Session type the command: **show vpn-sessiondb detail anyconnect** to view the details and statistics.

```

> show vpn-sessiondb detail anyconnect

Session Type: AnyConnect Detailed

Username      : harry                      Index      : 1
Assigned IP   : 198.19.10.57              Public IP   : 198.18.133.23
Protocol      : AnyConnect-Parent SSL-Tunnel DTLS-Tunnel
License       : AnyConnect Premium
Encryption    : AnyConnect-Parent: (1)none SSL-Tunnel: (1)AES-GCM-256 DTLS-Tunnel: (1)SHA1
Hashing       : AnyConnect-Parent: (1)none SSL-Tunnel: (1)SHA384 DTLS-Tunnel: (1)SHA1
Bytes Tx      : 17144589                  Bytes Rx    : 1234659
Pkts Tx       : 10931                    Pkts Rx     : 10618
Pkts Tx Drop  : 0                       Pkts Rx Drop : 0
Group Policy  : DfltGrpPolicy             Tunnel Group : RAVPN
Login Time    : 17:51:07 UTC Mon Jan 13 2020
Duration      : 0h:14m:23s
Inactivity    : 0h:00m:00s
VLAN Mapping  : N/A                       VLAN         : none
Auth Sess ID  : c61285510000100056elcae0b
Security Grp  : none                       Tunnel Zone  : 0

AnyConnect-Parent Tunnels: 1
SSL-Tunnel Tunnels: 1
DTLS-Tunnel Tunnels: 1

AnyConnect-Parent:
Tunnel ID     : 1.1
Assigned IP    : 198.18.133.23
Public IP     : 198.18.133.23
Encryption    : none                      Hashing      : none
TCP Src Port  : 51086                     TCP Dst Port : 443
Auth Mode     : userPassword
Idle Time Out : 30 Minutes                 Idle TO Left : 15 Minutes
Client OS     : win
Client OS Ver : 6.1.7601 Service Pack 1
Client Type   : AnyConnect
Client Ver    : Cisco AnyConnect VPN Agent for Windows 4.7.02036
Bytes Tx      : 3236293                    Bytes Rx     : 605
Pkts Tx       : 6                         Pkts Rx      : 0
Pkts Tx Drop  : 0                         Pkts Rx Drop : 0

SSL-Tunnel:
Tunnel ID     : 1.2
Assigned IP    : 198.19.10.57              Public IP     : 198.18.133.23
Encryption    : AES-GCM-256               Hashing       : SHA384
CipherSuite    : ECDHE-RSA-AES256-GCM-SHA384
Encapsulation  : TLSv1.2                  TCP Src Port  : 51093
TCP Dst Port  : 443                       Auth Mode     : userPassword
Idle Time Out : 30 Minutes                 Idle TO Left  : 15 Minutes
Client OS     : Windows
Client Type    : SSL VPN Client
Client Ver    : Cisco AnyConnect VPN Agent for Windows 4.7.02036
Bytes Tx      : 8105                       Bytes Rx      : 765
Pkts Tx       : 6                         Pkts Rx       : 11
Pkts Tx Drop  : 0                         Pkts Rx Drop  : 0
Filter Name    : #ACSAcl#-IP-PERMIT_ALL_TRAFFIC-57f6bd03

DTLS-Tunnel:
Tunnel ID     : 1.3
Assigned IP    : 198.19.10.57              Public IP     : 198.18.133.23
Encryption    : AES256                    Hashing       : SHA1
CipherSuite    : DHE-RSA-AES256-SHA
Encapsulation  : DTLSv1.0                 UDP Src Port  : 51968
UDP Dst Port  : 443                       Auth Mode     : userPassword
Idle Time Out : 30 Minutes                 Idle TO Left  : 29 Minutes
Client OS     : Windows
Client Type    : DTLS VPN Client
Client Ver    : Cisco AnyConnect VPN Agent for Windows 4.7.02036
Bytes Tx      : 13908201                   Bytes Rx      : 1233289
Pkts Tx       : 10919                     Pkts Rx       : 10607
Pkts Tx Drop  : 0                         Pkts Rx Drop  : 0
Filter Name    : #ACSAcl#-IP-PERMIT_ALL_TRAFFIC-57f6bd03

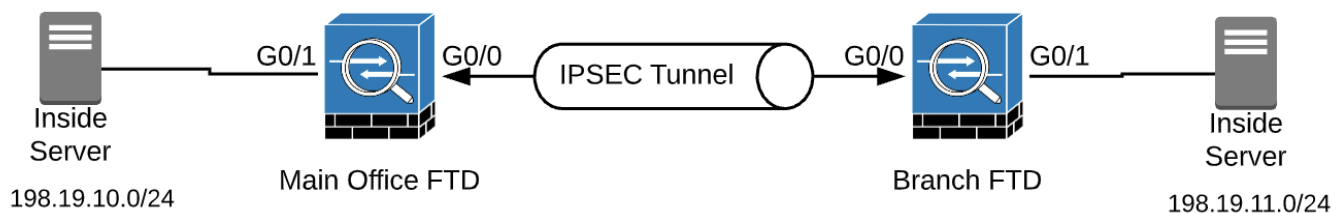
```

Scenario 2. Secure Site Connectivity between Main and Branch Office

This exercise consists of the following tasks.

- Create and verify objects needed for this lab exercise
- Configure site-to-site VPN
- Create NAT exemption
- Modify the access control policy and deploy changes
- Deploy the changes and test the configuration

The objective of this exercise is to configure a Site-to-Site IKEv2 VPN tunnel between two FMC Controlled NGFWs using pre-shared-keys. The Main Office NGFW is protecting the local LAN network 198.19.10.0/24 and the Branch Office NGFW is protecting the local LAN network 198.19.11.0/24. At the end of this exercise a server behind Main Office NGFW should be able to access a server behind the Branch Office NGFW and vice-versa.



Steps

Create network objects needed for this lab exercise

1. Navigate to **Objects > Object Management**. The **Network** object page will be selected.
 - a. Select **Add Network > Add Object**.
 - b. For Name, enter **MainOfficeNetwork**.
 - c. Select the **Network** radio button
 - d. Enter **198.19.10.0/24**.
 - e. Click **Save**.
2. Click **Add Network > Add Object**.
 - a. For Name, enter **Branch1OfficeNetwork**.
 - b. Click the **Network** radio button
 - c. Enter **198.19.11.0/24**.

d. Click **Save**.

Network

Add Network

A network object represents one or more IP addresses. Network objects are used in various places, including access control policies, network variables, intrusion rules, identity rules, network discovery rules, event s

Name	Value	Type
Branch1OfficeNetwork	198.19.11.0/24	Network
MainOfficeNetwork	198.19.10.0/24	Network

Run the Site to Site VPN Wizard

Here both the FTDs are managed by the FMC so the wizard can prepopulate the device information. However, it is also possible to configure and establish a Site to Site tunnel between an FMC managed FTD and FDM managed FTD. This is depicted in the FDM VPN lab guide. There are three kinds of VPN topologies available in the FMC Site to Site VPN wizard:

- Point to Point – This is the deployment method of choice where we need to establish a VPN tunnel between two endpoints.
- Hub and Spoke – This is the deployment method of choice for a use case involving one HQ/Main office and multiple Remote/Branch locations. It supports adding Spokes as **Extranet** if the Remote VPN endpoints are not managed by FMC.
- Full Mesh – This is the deployment method of choice when multiple VPN endpoints have tunnel connectivity to each other.

1. Navigate to **Devices > VPN > Site To Site**. Click **Add VPN > Firepower Threat Defense Device**.

NOTE: The other VPN choice, **Firepower Device**, is for configuring secure tunnels between Legacy Firepower devices and is not covered in this lab guide.

2. For Name enter **S2S_Branch1**.

- a. Confirm that for Network Topology, **Point to Point** is selected. Confirm that for IKE Version, **IKEv1** is **not checked**, and **IKEv2** is **checked**.

Node B

There are no VPN Topologies. Create a topology by adding [Firepower Device](#) (or) [Firepower Threat Defense Device](#).

Create New VPN Topology

Topology Name:*

S2S_Branch1

Network Topology:

Point to Point

Hub and Spoke

Full Mesh

IKE Version:*

☐ IKEv1
 ☒ IKEv2

Endpoints

IKE

IPsec

Advanced

Node A:

+

Device Name	VPN Interface	Protected Networks

Node B:

+

Device Name	VPN Interface	Protected Networks

Ensure the protected networks are allowed by access control policy of each device.

Cancel

Save

3. Click the **+** icon to the right of **Node A**. Fill out as in the figure below, and then click **OK**.

Node A

Node B

There are no VPN Topologies. Create a topology by adding [Firepower Device](#) (or) [Firepower Threat Defense Device](#).

Create New VPN Topology

Add Endpoint

Topology Name: NGFW1

Interface: outside

IP Address: 198.18.133.81

☐ This IP is Private

Connection Type: Bidirectional

Certificate Map:

Protected Networks:

☒ Subnet / IP Address (Network) ☐ Access List (Extended)

Network Objects

Available Networks

Q Search

ISE_Server

Lab_Networks

LAN_Network

MainOfficeNetwork

wwwin

wwwout

Selected Networks

MainOfficeNetwork

Cancel OK

Cancel OK

Cancel Save

1 Ensure the protected networks are allowed by access control policy of each device.

4. Click the **+** icon to the right of **Node B**. Fill out the fields in the figure below, then click **OK**.

Node A

Node B

There are no VPN Topologies. Create a topology by adding [Firepower Device](#) (or) [Firepower Threat Defense Device](#).

Create New VPN Topology

Add Endpoint

Topology Name: NGFWBR1

Interface: outside

IP Address: 198.18.128.81

☐ This IP is Private

Connection Type: Bidirectional

Certificate Map:

Protected Networks:

☒ Subnet / IP Address (Network) ☐ Access List (Extended)

Network Objects

Available Networks

Q Search

any

any-ipv4

any-ipv6

Branch1OfficeNetwork

DNS_Server

Extranet129

Selected Networks

Branch1OfficeNetwork

Cancel OK

Cancel OK

Cancel Save

1 Ensure the protected networks are allowed by access control policy of each device.

5. Select the **IKE** tab.

- a. Under IKEv2 Settings, for Policy, select **AES-SHA-SHA**.
- b. Under IKEv2 Settings, for Authentication Type, select **Pre-shared Automatic Key**.

NOTE: The Automatic setting can only be used if the FMC is managing both endpoints. In this case, the FMC can generate a random shared key on both the endpoints it is managing.

Create New VPN Topology

Topology Name:*
S2S_Branch1

Network Topology:

IKE Version:*
☐ IKEv1
 ☒ IKEv2

Endpoints

IKEv1 Settings

Policy:* preshared_sha_aes256_dh5_5 +

Authentication Type: Pre-shared Automatic Key

Pre-shared Key Length:* 24 Characters (Range 1-127)

IKEv2 Settings

Policy:* AES-SHA-SHA +

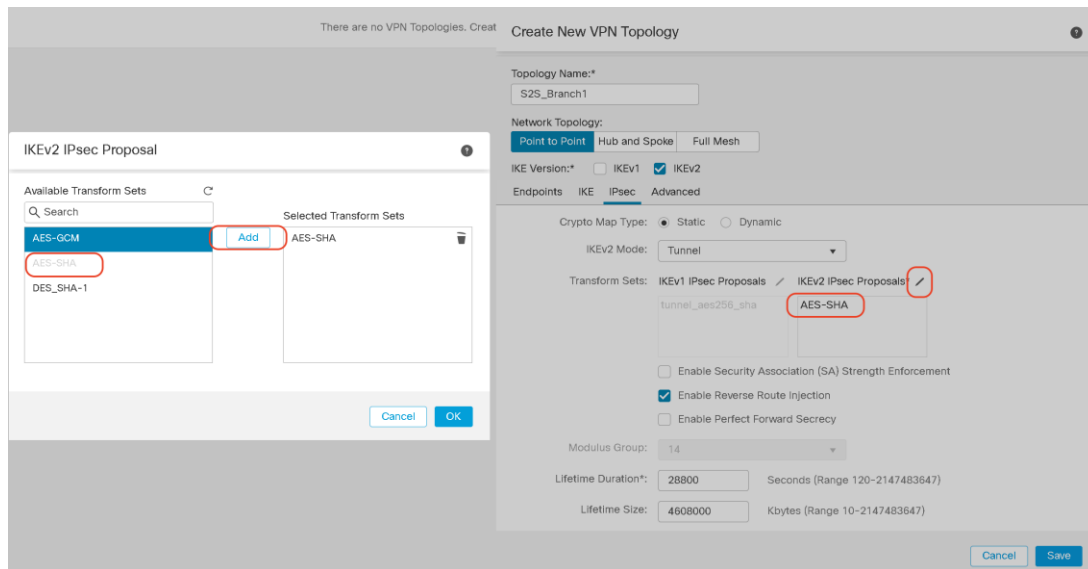
Authentication Type: Pre-shared Automatic Key

Pre-shared Key Length:* 24 Characters (Range 1-127)

Cancel Save

6. Select the **IPsec** tab, change the IKEv2 IPsec Proposal to **AES-SHA**.

- a. Click on the **pencil** (Edit) icon by IKEv2 IPsec Proposal.
- b. Click on **AES-SHA** and Click **Add**.
- c. Delete **AES-GCM**.



NOTE: The user can define their own IKE and IPSEC Policies. However, for the purpose of this lab we are using pre-defined set of policies. You can view the details of these policies under the Object Management page at **Objects > VPN**. Also, there are more settings available under the **Advanced** tab such as Keepalives, fragmentation, NAT-T etc. For the purpose of this lab, we are using default configuration of those settings.

7. Click **OK** and **Save**.

Configuration to allow VPN Traffic at Main Office

In this section, we will configure the Access-Control Policy at the Main Office FTD (NGFW1) to allow inbound traffic from the Branch FTD network. Also, we will configure NAT exemption rule for this traffic.

Create NAT exemption at Main Office

NOTE: NAT exemption is used so that the addresses are not translated by NAT. To do this you need to have the packets translated by the NAT process back to their original addresses. This must be done before any other NAT statements so you will put the rule in the NAT Rules Before Category. This will be done in NAT policies for both VPN endpoints – NGFW1 and NGFWBR1. Details around the types of NAT can be found [here](#).

1. Navigate to **Devices > NAT**.
2. Click the pencil icon to edit the **Default NAT** policy.
3. Click **Add Rule**.
 - a. In **Category** and **NAT Rules Before** from the **Insert** drop-down list should be selected by default. Under **Interface Objects** tab.

- i. Select **InZone1** and click **Add to Source**
 - ii. Select **OutZone** and click **Add to Destination**.
- b. Select the **Translation** tab.
 - i. Select **MainOfficeNetwork** from the Original Source drop-down list.
 - ii. Select **MainOfficeNetwork** from the Translated Source drop-down list.
 - iii. Select **Branch1OfficeNetwork** from the Original Destination drop-down list.
 - iv. Select **Branch1OfficeNetwork** from the Translated Destination drop-down list.
- c. You will notice that in the above translation rule the original and translated networks are the same for source and destination networks. This is called Identity NAT.

Add NAT Rule

NAT Rule: Manual NAT Rule

Insert: In Category NAT Rules Before

Type: Static

☒ Enable

Description:

Interface Objects **Translation** PAT Pool Advanced

Original Packet	Translated Packet
Original Source:* MainOfficeNetwork	Translated Source: Address
Original Destination: Address	Translated Destination: MainOfficeNetwork
Original Source Port:	Translated Source Port:
Original Destination Port:	Translated Destination Port:

Cancel OK

4. Click **Advanced** and Check **Do not proxy ARP on Destination Interface**. Click **OK**.

Add NAT Rule

NAT Rule: Manual NAT Rule

Insert: In Category NAT Rules Before

Type: Static

☒ Enable

Description:

Interface Objects Translation PAT Pool **Advanced**

☐ Translate DNS replies that match this rule
☐ Falthrough to Interface PAT(Destination Interface)
☐ IPv6
☐ Net to Net Mapping
☒ Do not proxy ARP on Destination Interface
☐ Perform Route Lookup for Destination Interface
☐ Unidirectional

Cancel OK

NOTE: Enabling Do not proxy ARP on Destination Interface is critical in this lab exercise. If you miss this step, the NGFW1 will start replying to ARP requests received on the outside interface, for IP addresses which are part of the NGFW1 outside interface subnet. This might cause your pod to have access issues, since all devices are managed in band.

5. Click **Save**.

Modify the access control policy on Main office

You will now create a rule on the Main Office NGFW to allow traffic initiated from the Branch office LAN Network to enter the Main Office LAN network over the Site to Site VPN tunnel. Since, the NGFW1 (HA_TEST In case HA scenario was completed) is protecting the Main Office and the traffic flow on NGFW1 (HA_TEST in case HA was completed) is controlled by the **Base_Policy** or **NGFW_Policy** Access Control Policy, we will be making changes to it.

1. Navigate to **Policies > Access Control**. Click the **pencil** icon to edit the **Base_Policy**.
2. Click **Add Rule**.
 - a. Call the rule **S2S_Branch1_VPN_Access**.
3. Select **into Default** from the **Insert** drop-down list. This will become the last rule in the access control policy.
4. Keep the action at **Allow**.
5. The **Zones** tab should already be selected.
 - a. Select **OutZone**, and click **Add to Source**.
 - b. Select **InZone1** and click Add to **Destination**.

The screenshot shows the 'Add Rule' configuration interface. At the top, the rule name is 'S2S_Branch1_VPN_Access', it is 'Enabled', and the 'Insert' dropdown is set to 'into Default'. The 'Action' is 'Allow'. Below this, the 'Zones' tab is active, displaying a list of 'Available Zones' (InZone, OutZone, GRE). 'OutZone' is selected and added to the 'Source Zones (1)' list, and 'InZone' is selected and added to the 'Destination Zones (1)' list. The 'Add' button is located at the bottom right of the interface.

6. Select the **Networks** tab:

- a. Select **Branch1OfficeNetwork** and click **Add to Source**.
- b. Select **MainOfficeNetwork** and click **Add to Destination**.

7. Select the **Inspection** tab.

- a. Select Demo Intrusion Policy from the Intrusion Policy drop-down list.
- b. Select **Demo File Policy** from the **File Policy** drop-down list.

8. Click **Add** to add this rule to the access control policy.

9. Click **Save** to save the access control policy.

Configuration to allow VPN Traffic at Branch Office

In this section, we will configure the Access-Control Policy at the Branch Office FTD (NGFWBR1) to allow inbound traffic from the Main FTD network. Also, we will configure NAT exemption rule for this traffic.

Create NAT exemption for Branch1

1. Navigate to **Devices > NAT**.
2. Click the pencil icon to edit the **Branch NAT policy**
3. Click **Add Rule**.
 - a. Leave **In Category** and **NAT Rules Before** from the NAT Rule drop-down list selected. Under **Interface Objects**:

- i. Click **InZone** and **Add to Source**.
- ii. Click **OutZone** and **Add to Destination**.

Add NAT Rule

NAT Rule:
Manual NAT Rule

Insert:
In Category NAT Rules Before

Type:
Static

☒ Enable

Description:

Interface Objects Translation PAT Pool Advanced

Available Interface Ob...
Q Search by name

InZone
OutZone

Add to Source
Add to Destination

Source Interface Objects (1)
InZone

Destination Interface Obj... (1)
OutZone

Cancel OK

4. Click on the **Translation** tab.
 - a. Select **Branch1OfficeNetwork** from the Original Source drop-down list.
 - b. Select **Branch1OfficeNetwork** from the Translated Source drop-down list.
 - c. Select **MainOfficeNetwork** from the Original Destination drop-down list.
 - d. Select **MainOfficeNetwork** from the Translated Destination drop-down list.

Add NAT Rule

NAT Rule: Manual NAT Rule

Insert: In Category NAT Rules Before

Type: Static

☒ Enable

Description:

Interface Objects Translation PAT Pool Advanced

Original Packet Translated Packet

Original Source:* Branch1OfficeNetwork

Original Destination: Address MainOfficeNetwork

Original Source Port:

Original Destination Port:

Translated Source: Address Branch1OfficeNetwork

Translated Destination: MainOfficeNetwork

Translated Source Port:

Translated Destination Port:

Cancel OK

5. Click **Advanced** and Select **Do not proxy ARP on Destination Interface**. Click **OK**

Add NAT Rule

NAT Rule: Manual NAT Rule

Insert: In Category NAT Rules Before

Type: Static

☒ Enable

Description:

Interface Objects Translation PAT Pool **Advanced**

☐ Translate DNS replies that match this rule

☐ Falthrough to Interface PAT(Destination Interface)

☐ IPv6

☐ Net to Net Mapping

☒ Do not proxy ARP on Destination Interface

☐ Perform Route Lookup for Destination Interface

☐ Unidirectional

Cancel OK

NOTE: Enabling Do not proxy ARP on Destination Interface is critical in this lab exercise. If you miss this step, the NGFWBR1 will start replying to ARP requests received on the outside interface, for IP addresses which are part of the NGFWBR1 outside interface subnet. This might cause your pod to have access issues, since all devices are managed in band.

6. Click **Save** to save the NAT policy.

Modify the Access Control Policy on Branch

You will now create a rule on the Branch Office NGFW to allow traffic initiated from the Main office LAN Network to enter the Branch Office LAN network over the Site to Site VPN tunnel. Since, the NGFWBR is protecting the Branch Office and the traffic flow on NGFWBR1 is controlled by the **Branch Access Control Policy** Access Control Policy, we will be making changes to it.

1. Now Modify the **Branch Access Control Policy** as below to allow inbound connections
2. Click **Add Rule**.
 - a. Call the rule **S2S_Main_VPN_Access**.
3. Select **into Default** from the Insert drop-down list. This will become the last rule in the access control policy.
4. Keep the action at **Allow**.
5. Under the **Zones** tab:
 - a. Select **OutZone** and click **Add to Source**.
 - b. Select **InZone** and click **Add to Destination**.
6. Select the **Networks** tab, select **Branch1OfficeNetwork**, and click **Add to Destination**.
7. Select the **Networks** tab, select **MainOfficeNetwork**, and click **Add to Source**.
8. Select the **Inspection** tab.
 - a. Select **Demo Intrusion Policy** from the Intrusion Policy drop-down list.
 - b. Select **Demo File Policy** from the File Policy drop-down list.
9. Click **Add** to add this rule to the access control policy.
10. Click **Save** to save the access control policy.

Deploy the changes and test the configuration

Deploy the Changes

1. Deploy the changes on the FMC and wait for the deployment to complete.
2. Select **NGFW1** and **NGFWBR1**. Click **Deploy**.

NOTE: You will get a warning during deployment. The deployment warning is due to the use of DH groups 5 and 24 in the IKEv2 Policy. DH Groups 2,5 and 24 have been deprecated in version 6.5 and in later versions they will not be supported.

3. Click **Proceed**. Wait for the deployment to complete. You can monitor the status of the deployment by clicking on the **Notifications** tab.

Test the VPN connectivity

1. From jumpbox, open PUTTY and login to NGFW1 and NGFWBR1 with credentials – User: **admin** Password: **C1sco12345**
2. From the **NGFW1** CLI, type **show crypto ipsec sa peer 198.18.128.81**. There should be no IPSec security associations.
3. From the **NGFWBR1** CLI, type **show crypto ipsec sa peer 198.18.133.81**. There should be IPSec security associations.
4. Open a PUTTY Session to **Inside Linux Server** Login: **root** Password: **C1sco12345**.
 - a. Type the command : **sudo route del -net 198.19.11.0 gw 198.19.10.2 netmask 255.255.255.0 dev eth0**

Note: The Inside Linux server already has a route for the Branch1 Office Network with the next hop set as NGFW2. The above command is used to point the route for the Branch1 Office Network via the NGFW1.

5. From the Inside Linux server CLI, type **ping branch**. Wait a few seconds, and the ping should succeed.

```

root@inside ~]# ping branch
PING branch (198.19.11.200) 56(84) bytes of data:
 4 bytes from branch (198.19.11.200): icmp_seq=2 ttl=64 time=48.0 ms
 4 bytes from branch (198.19.11.200): icmp_seq=3 ttl=64 time=54.4 ms
 4 bytes from branch (198.19.11.200): icmp_seq=4 ttl=64 time=56.1 ms
 4 bytes from branch (198.19.11.200): icmp_seq=5 ttl=64 time=54.3 ms
 4 bytes from branch (198.19.11.200): icmp_seq=6 ttl=64 time=24.4 ms
 4 bytes from branch (198.19.11.200): icmp_seq=7 ttl=64 time=33.8 ms
 4 bytes from branch (198.19.11.200): icmp_seq=8 ttl=64 time=64.7 ms
 4 bytes from branch (198.19.11.200): icmp_seq=9 ttl=64 time=43.9 ms
 4 bytes from branch (198.19.11.200): icmp_seq=10 ttl=64 time=54.4 ms
 4 bytes from branch (198.19.11.200): icmp_seq=11 ttl=64 time=46.4 ms
 4 bytes from branch (198.19.11.200): icmp_seq=12 ttl=64 time=54.3 ms
 4 bytes from branch (198.19.11.200): icmp_seq=13 ttl=64 time=44.2 ms
 4 bytes from branch (198.19.11.200): icmp_seq=14 ttl=64 time=54.7 ms
 4 bytes from branch (198.19.11.200): icmp_seq=15 ttl=64 time=44.1 ms
C
-- branch ping statistics ---
5 packets transmitted, 14 received, 6% packet loss, time 14021ms

```

6. From the NGFW1 CLI, type **show crypto ipsec sa**. There should now be an IPSec security association.

```

> show crypto ipsec sa
Interface: ISP-Side
Crypto map tag: CSM_ISP-Side_map, seq num: 1, local addr: 198.18.133.2

access-list CSM_IPSEC_ACL_1 extended permit ip 198.19.10.0 255.255.255.0 198.19.11.0 255.255.255.0
local ident (addr/mask/prot/port): (198.19.10.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (198.19.11.0/255.255.255.0/0/0)
current_peer: 198.18.133.142

#pkts encaps: 9, #pkts encrypt: 9, #pkts digest: 9
#pkts decaps: 9, #pkts decrypt: 9, #pkts verify: 9
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 9, #pkts comp failed: 0, #pkts decomp failed: 0
#pre-frag successes: 0, #pre-frag failures: 0, #fragments created: 0
#PMTUs sent: 0, #PMTUs rcvd: 0, #decapsulated frgs needing reassembly: 0
#TFC rcvd: 0, #TFC sent: 0
#Valid ICMP Errors rcvd: 0, #Invalid ICMP Errors rcvd: 0
#send errors: 0, #recv errors: 0

local crypto endpt.: 198.18.133.2/500, remote crypto endpt.: 198.18.133.142/500
path mtu 1500, ipsec overhead 58(36), media mtu 1500
PMTU time remaining (sec): 0, DF policy: copy-df

```

7. On the Jump desktop, open the PUTTY link. Double click on the preconfigured session called **Branch Linux Server**.
8. Login as **root**, password **C1sco12345**
9. Type **curl inside**. This should succeed with a similar output as below

```

[root@branch ~]# curl inside
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html><head><title>Cisco&#8482; Technical Marketing</title>

<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1">

<link rel="stylesheet" type="text/css" media="screen" href="css/common.css">
<link rel="stylesheet" type="text/css" media="screen" href="css/subnav.css">
<link rel="stylesheet" type="text/css" media="print" href="css/print.css">
</head>

<body class="training">

  <a href="/index.html"></a>
<hr>
<h1> inside.dcloud.local (198.19.10.200) Website </h1>
<hr>
<a href=https://inside.dcloud.local><h1>Go to secure server</h1></a>
<hr>
<!-- content // start -->

<h1><a href="files">Files</a></h1>
<h1><a href="certs">Certs</a></h1>

<hr>
<p style="position: absolute; right: 15px; font-size: 90%; line-height: 300%;" >
September 2018
</p>

<!-- content // end -->
</body></html>
[root@branch ~]#

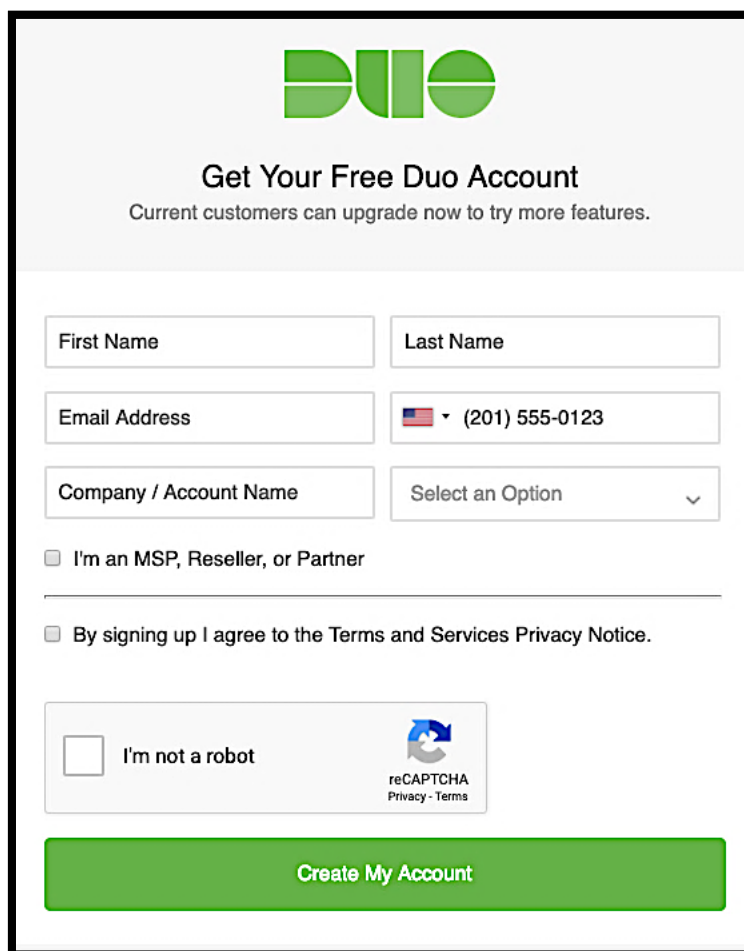
```

Appendix A. DUO Account Setup

Create Duo Account and enroll your mobile device

NOTE: You can skip this section and continue to the Add Firepower application and enroll VPN user to Duo Account section if you already have a Duo Account and have your mobile device registered to this account.

1. Since we will be using Duo as a factor in our RA VPN authentication, let's first setup a Duo Account for ourselves. Duo allows you to create a free trial account and link it to your mobile device. This will be sufficient for our lab exercise.
2. Navigate to <https://duo.com/docs/getting-started> and follow steps 1-4 under the **Getting Started** section.



The screenshot shows the Duo account creation page. At the top is the Duo logo. Below it, the heading 'Get Your Free Duo Account' is displayed, followed by the text 'Current customers can upgrade now to try more features.' The form contains several input fields: 'First Name', 'Last Name', 'Email Address', and a phone number field with a dropdown menu showing '(201) 555-0123'. There is also a 'Company / Account Name' field and a 'Select an Option' dropdown. Below these fields are two checkboxes: 'I'm an MSP, Reseller, or Partner' and 'By signing up I agree to the Terms and Services Privacy Notice.' At the bottom left is a checkbox labeled 'I'm not a robot' next to a reCAPTCHA widget. A large green button labeled 'Create My Account' is at the bottom.

1. Once you are done, we can proceed to the next section to add our test VPN user and the Firepower application to your account.

Add Firepower application and enroll VPN user to Duo Account

1. Log in to the **Duo Admin Panel** [https://admin.duosecurity.com] and navigate to **Applications**.
2. Locate **Cisco Firepower Threat Defense VPN** in the applications list

Dashboard > Applications > Protect an Application

Protect an Application

Add an application that you'd like to protect with Duo two-factor authentication.
You can start with a small "proof-of-concept" installation — it takes just a few minutes, and you're the only one that will see it, until you decide to add others.

Documentation: [Getting Started](#)

Choose an application below to get started.

**Cisco Firepower Threat Defense VPN**[Protect this Application](#) | [Read the documentation](#)

3. Click **Protect this Application** to get your integration key, secret key, and API hostname

Dashboard > Applications > Cisco Firepower Threat Defense VPN

Cisco Firepower Threat Defense VPN

[Authentication Log](#) | [Remove Application](#)

Follow the [Cisco Firepower Threat Defense \(FTD\) Remote Access VPN application instructions](#).

[Reset Secret Key](#)

Details

Integration key	DIBS6UGGW1KH44Y7LJCH	select
Secret key	Click to view.	select
Don't write down your secret key or share it with anyone.		
API hostname	api-29acc416.duosecurity.com	select

NOTE: Treat your secret key like a password. The security of your Duo application is tied to the security of your secret key (skey). Secure it as you would any sensitive credential. Don't share it with unauthorized individuals or email it to anyone under any circumstances

4. There is a lot more functionality associated with the application such as Policy, but this is out of the scope of the lab.
5. Let's now add our VPN user account that will be used for testing. Navigate to **Users** on the **Duo Admin Panel** and click on **Add User**. Type in **vpnuser** for the **Username** field and click on **Add User**.

Dashboard > Users > Add User

Add User

Adding Users
Most applications allow users to enroll themselves after they complete primary authentication.
[Learn more about adding users](#)

Username
Should match the primary authentication username.

Add User

NOTE: If you cannot receive SMS messages for enrollment, follow step #6, otherwise follow from step #7 onwards.

6. In the **vpnuser** dashboard add in an email address that you can use for enrollment. Click on **Save Changes**. Now you can click on the **Send Enrollment Email** link to get a message that contains a link that you can use to add a phone or other 2FA authentication device. Jump to **step #12**.

Dashboard > Users > vpnuser

vpnuser

Logs | Send Enrollment Email | Send to Trash

This user has not enrolled yet. See our [enrollment documentation](#) to learn more about enrolling users.

7. In the **vpnuser** dashboard add in your phone number so that an enrollment message can be sent. Scroll down on the new user's details page to the **Phones** table and click on **Add Phone**

Phones

To use two-factor with a specific phone, type 'push1', 'sms1', 'phone2', 'sms2', etc, as your factor. You may rearrange the phones by dragging and dropping in the table.

[Add Phone](#)

This user has no phones. [Add one.](#)

8. Chose Phone, and type in the phone number (leave this field blank if adding a tablet). Click the Add Phone button.

[Dashboard](#) > [Users](#) > [avandalay](#) > Add Phone

Add Phone

Type



Phone



Tablet

Phone number



+1 734-555-8008

[Show extension field](#)

[Add Phone](#)

9. Choose the appropriate phone Type and Platform from the drop-down menus and enter a Device name (this field can be left blank). If you know the device is a smartphone but aren't sure exactly what the platform is, choose Generic Smartphone and the actual platform will be set when the user completes activation. Click the Save Changes button.

Settings

Number	 +1 734-555-8008	Show extension settings
Device name	 Optional. Examples: "Work phone", "Old iPod touch"	
Type	Mobile	
Platform	iOS	
Save Changes		

10. Click the Activate Duo Mobile link in the Device Info section. This link is only available when you set the phone type to Mobile and selected something other than Unknown as the platform. Then on the next page click the Generate Duo Mobile Activation Code button.

Device Info



Not using Duo Mobile
[Activate Duo Mobile](#)

Activate Duo Mobile

This form allows you to generate a new activation code for this phone's Duo Mobile application. The Duo Mobile application allows the user to generate passcodes on their mobile device or authenticate via Duo Push.

Note: Generating an activation code will invalidate any existing Duo Mobile credentials for this device until it is activated with the new activation code.

Phone 734-555-4211

Expiration 24 hours after generation

Generate Duo Mobile Activation Code

- Next, you'll see two text messages that you can send. The first has a link that helps the user install Duo Mobile. The second message has a code that the user can use to immediately add the account to their Duo Mobile app. Click the Send Instructions by SMS button to send the text messages to the user's phone.

Activate Duo Mobile

A new Duo Mobile activation code has been generated, and any old credentials have been invalidated. You now have the option to deliver the installation instructions and/or activation instructions to the user by SMS.

Phone 734-555-4211

Installation Instructions ☒ Send installation instructions via SMS

Welcome to Duo! Please install Duo Mobile from your app store.

Activation Instructions ☒ Send activation instructions via SMS

To activate the app, tap and open this link with Duo Mobile: <https://m-xxxxxxxxx.duosecurity.com/activate/LSdDPKMB2312HktaAKcll>

Send Instructions by SMS or skip this step

- You would have received an email message or an SMS message with an activation link and QR code, as well as links to the Duo Mobile app for all supported platforms. You need to either open the link on your device or scan the QR code with the Duo Mobile App to add the Duo account and activate vpnuser. Once done, the Duo account is added, and vpnuser is fully activated.

Appendix B. FMC Pre-configuration

After the initial installation, several configuration steps were performed on the FMC to expedite the lab exercises. These configuration steps are detailed in this appendix.

- Configuration A1,1 : Objects
- Configuration A1,2: Demo file policy
- Configuration A1,3: Demo intrusion policy
- Configuration A1,4: Demo SSL policy
- Configuration A1,5: Custom detection list
- Configuration A1,6: Add resetapiuser.
- Configuration A1,7: Install server certificate

Steps

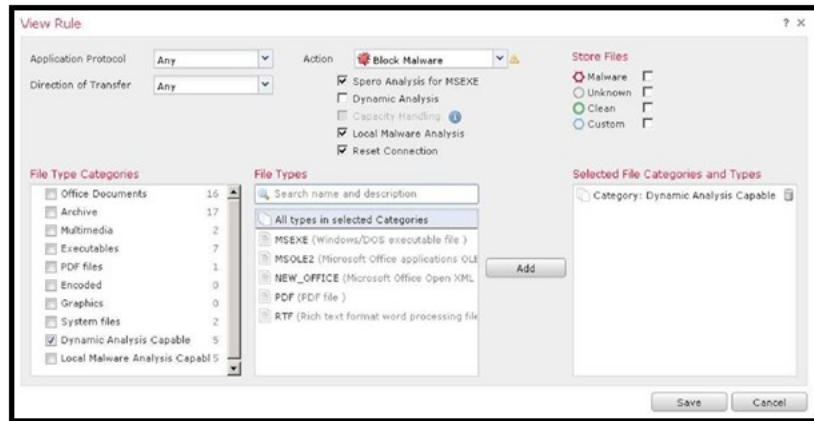
Configuration A1,1: Network Objects

Name	Type	Network
VPNPoolIPs	Range	198.19.10.57-198.19.10.62
LAN_Network	Network	198.19.10.0/24
ISE_Server	Host	198.19.10.130
DNS_Server	Network	198.19.10.100

Configuration A1,2: Demo file policy

1. **Navigate to** Policies > Access Control > Malware & File.
2. **Click** New File Policy. **Enter a name** Demo File Policy. **Click** Save.
3. Click **Add File Rule**. This rule will block malware found in files MSEXE, MSOLE2, NEW_OFFICE and PDFs.
4. **For Action select** Block Malware.
5. Check the Spero and **Local Malware Analysis** checkboxes.
6. Under **File Type Categories**, check *Dynamic Analysis Capable*. **Note** that several file types belong to this category. Click **Add**.

7. Your screen should look like the figure below.



8. Click **Save**. Ignore the warning and click **OK**, when prompted.
9. Click **Add File Rule**. This rule will block RIFF files. You will use an AVI file to test this rule, since an AVI file is a type of RIFF file. But **Note** that AVI is not listed separately as a file type.
10. For Action **select** Block Files.
11. Under **File Types**, type **riff** into the search box. Select **RIFF** from the list. Click **Add**.
12. Use default values for other settings. Your screen should look like the figure below.
13. Click **Save**.



NOTE: You cannot change the order of the rules you create. The order of the rules does not matter. The action of the rule determines its precedence. The precedence of actions is as follows.

- Block Files
- Block Malware
- Malware Cloud Lookup

- Detect Files

14. Select the **Advanced** tab. Confirm that **Enable Custom Detection List** is selected.
15. Check the **Inspect Archives** checkbox.



NOTE: Archives unable to be inspected are corrupt archive, or archives with a depth that exceeds the Max Archive Depth.

16. Click the **Save** button in the upper-right to save the file policy.

Configuration A1,3: Demo intrusion policy

1. **Navigate to** Objects > Intrusion Rules. **Click** Import Rules.
 - a. **Select the** Rule update or text rule file to upload and install **radio button**.
 - b. **Click Browse** and open the **Snort_Rules.txt** file in the **Files** folder of the Jump desktop.

NOTE: This file contains 2 simple Snort rules that are useful for testing IPS. They do not resemble published snort rules.

```
alert tcp any any -> any any (msg:"ProjectQ replaced"; content:"ProjectQ"; replace:"ProjectR"; sid: 1001001; rev:1;)
alert tcp any any -> any any (msg:"ProjectZ detected"; content:"ProjectZ"; sid: 1001002; rev:1;)
```

The first rule replaces the string ProjectQ with ProjectR. The second detects the string ProjectZ. Since the rules do not specify where the string is in the flow, they could cause issues in a production deployment.

- c. **Click Import**. The import process will take a minute or two. When it completes you will see the **Rule Update Import Log page**. Confirm that 2 rules were successfully imported.
2. **Navigate to** Policies > Access Control > Intrusion.
3. **Click** Create Policy.
 - a. **Set Name to** Demo Intrusion Policy.
 - b. Make sure that **Drop when Inline** is checked.
 - c. Select Balanced Security and Connectivity as **Base Policy**.

d. Click **Create and Edit Policy**.

4. You will now modify the rules states for this new policy.

- a. Click **Rules** under Policy Information menu on the left-hand side of the **Edit Policy** page.
- b. Select **local** from the Category section of the rules. You should see the 2 uploaded rules. The light green arrows on the right of each rule indicate that the rules are disabled for this policy.
- c. Check the checkbox next to the first rule. Select Generate Events from the Rule State drop-down menu. Click OK. Uncheck the checkbox next to the first rule.
- d. Check the checkbox next to the second rule. Select Drop and Generate Events from the Rule State drop-down menu. Click OK.
- e. Clear the filter by clicking on the X on the right side of the Filter text field.
- f. Select **SID** from the **Rule Content** section of the rules. Enter 336 into the **Enter the SID** filter popup. Click **OK**.
- g. Check the checkbox next to the rule. Select **Drop and Generate Events** from the **Rule State** drop-down menu. Click OK.

NOTE: This rule looks for a change to the root home directory in FTP traffic established on port 21. It only looks for traffic coming from the external network, but in our lab, we use the default value of \$EXTERNAL_NET, which is any, so the rule can be triggered in both directions.

An interesting exercise would be to modify this rule to search in FTP traffic in any direction, and to use the appid attribute to detect FTP traffic on any port.

Click **Policy Information** in the menu on the upper left.

Click **Commit Changes**.

Click **OK**.

Configuration A1,4: Demo SSL policy

1. **Navigate to** Objects > Object Management > PKI > Internal CAs.

- a. Click **Import CA**.
 - b. For **Name**, enter Verifraud.
 - c. Click the **Browse** button to the right of the text **Certificate Data or, choose a file**.
 - d. Browse to the **Certificates** folder on the Jump desktop.
 - e. **Upload** Verifraud_CA.cer.
 - f. Click the **Browse** button to the right of the text *Key or, choose a file*.
 - g. **Upload** Verifraud_CA.key.
 - h. Click **Save**.
2. You will exempt from decryption infrastructure devices, such as the FMC and AMP Private Cloud. To do this, create a network object that includes these devices.
 - a. **Navigate to** Objects > Object Management > Network.
 - b. **Click** Add Network > Add Object.
 - c. For **Name**, enter Infrastructure.
 - d. For Network, enter 198.19.10.80-198.19.10.130.
 - e. Click **Save** to save the network object.
3. Click the text **Add a new policy** or click the **New Policy** button.
 - a. For **Name**, enter Demo SSL Policy.
 - b. Leave the default action to **Do not decrypt**.
 - c. Click **Save**. Wait a few seconds, and the policy will open for editing.
4. Click **Add Rule**.
 - a. **For Name, enter** Exempt Infrastructure.
 - b. **Leave Action set to** Do Not decrypt.
 - c. **In the Networks tab, under Networks, select** Infrastructure, **and click** Add to Source.
 - d. Click **Add** to add this rule to the SSL policy.
5. Click **Add Rule**.
 - a. **For Name, enter** Decrypt Search Engines.
 - b. **Set Action to** Decrypt - Resign.
 - c. Select **Verifraud** from the drop-down list to the right of the word **with**.
 - d. In the **Applications** tab, under **Application Filters**, search for **Sear**. You will see **Search Engine** under **Categories**. Check this checkbox, and click **Add to Rule**.
 - e. Select the **Logging** tab and check the **Log at End of Connection** checkbox.
 - f. Click **Add** to add this rule to the SSL policy.

6. Click **Add Rule**.
 - a. For Name, enter **Decrypt Other**.
 - b. **Set Action to Decrypt - Resign**.
 - c. Select Verifraud from the drop-down list to the right of the word **with**.
 - d. Select the **Logging** tab and check the **Log at End of Connection** checkbox.
 - e. Click **Add** to add this rule to the SSL policy.
7. Click **Save** to save the SSL policy.

NOTE: The Replace Key checkbox deserves explanation. Whenever the action is set to Decrypt - Resign, Firepower will replace the public key. The Replace Key checkbox determines how the decrypt action is applied to self-signed server certificates.

If Replace Key is deselected, self-signed certificates are treated like any other server certificates. Firepower replaces the key and resigns the certificate. Generally, the endpoint is configured to trust Firepower, and therefore will trust this resigned certificate.

If Replace Key is selected, self-signed certificates are treated differently. Firepower replaces the key, and generates a new self-signed cert. The browser on the endpoint will generate a certificate warning.

In other words, checking the Replace Key checkbox makes the resign action preserve lack-of-trust for self-signed certificates.

Configuration A1,5: Custom detection list

There is a harmless file called Zombies.pdf that will trigger a malware event, assuming the cloud lookup succeeds. Sometimes labs have issues with cloud connectivity. Therefore, this is added to the custom detection list to ensure it will trigger a malware event.

1. **Navigate to** Objects > Object Management > File List.
2. Click the pencil icon to edit the **Custom-Detection-List**.
 - a. Select **Calculate SHA** from the **Add by** drop-down list.
 - b. Click **Browse**.
 - c. Browse to the Files folder on the Jump desktop.
 - d. Select **Zombies.pdf** and click **OK**.
 - e. **Click** Calculate and Add SHAs.
 - f. Click **Save**.

Configuration A1,6: Add restapiuser

It is convenient to have a separate use to use the API Explorer. This allows use of both the FMC and API Explorer at the same time.

1. **Navigate to** System > Users. **Click** Create User.

- a. For **User Name**, enter `restapiuser`.
- b. For **Password**, enter **C1sco12345** Confirm the password.
- c. Set Maximum Number of Failed Logins to 0.
- d. Check the **Administrator** checkbox.

Configuration A1,7: Install server certificate

By default, the FMC UI uses a self-signed certificate. This is replaced by a certificate signed by the pod AD server, which the Jump PC browsers trust.

1. **Navigate to** Objects > Object Management > PKI > Trusted CAs.
 - a. **Click** Add Trusted CA.
 - b. For **Name**, enter **dCloud**.
 - c. Click the **Browse** button to the right of the text **Certificate Data or, choose a file**.
 - d. Browse to the **Certificates\Lab Certificates\Other Certificates** folder on the Jump desktop.
 - e. **Upload** AD-ROOT-CA-CERT.cer.
 - f. Click **Save**. Connect to the FMC CLI via SSH. Become root by typing **sudo -i**. The Sudo password is **C1sco12345**
 - g. **Type** `cd /etc/ssl and then type cp server* /root`.
 - h. **Type** `cat > /etc/ssl/server.crt`
 - i. From the **Certificates\Lab Certificates\Other Certificates** folder on the Jump desktop edit the file **fmc.cer** with Notepad++.
 - j. Select all, and then copy and paste into the FMC CLI
 - k. Type **Ctrl+D**.
 - l. **Type** `cat > /etc/ssl/server.key`
 - m. From the **Certificates\Lab Certificates\Other Certificates** folder on the Jump desktop edit the file **fmc.key** with Notepad++.
 - n. Select all, and then copy and paste into the FMC CLI
 - o. Type **Ctrl+D**.
 - p. **Type** `pmtool restartbyid httpsd`.

Appendix C. FDM Pre-Configuration

Configuration A1,1: Objects

The following objects have been pre-configured.

Name	Type	Network
Corporate_LAN	Network	198.19.10.0/24
DNS_Server	Host	198.19.10.100
VPN_Pool	Network	198.19.10.64/29
ISE_Server	Host	198.19.10.130
DCloud_LAN1	Network	10.0.0.0/8
DCloud_LAN2	Network	198.19.255.0/24
DCloud_Jumpbox	Host	198.18.133.50

Appendix D. ISE RA VPN Configuration

ISE was configured to support all the lab exercises. In this appendix, this configuration is summarized. Note that there is an ISE link on the Firefox bookmarks toolbar. The credentials should prepopulate. They are username **admin**, password **C1sco12345**

NOTE: This appendix is not a tutorial on ISE. It does not go into details about how ISE is configured. It only covers the details required to configure RA VPN components for the lab exercises in this guide. The configurations are described in a top-down manner. To create this configuration, you would probably prefer to build these objects from the bottom-up.

Authorization policies

1. Navigate to **Policy > Authorization**. The first two policies were created for this lab: **AC-IT-Policy** and **AC-Default-Policy**. These policies reference two authorization profiles: AC-Auth-IT and AC-Auth-Default.

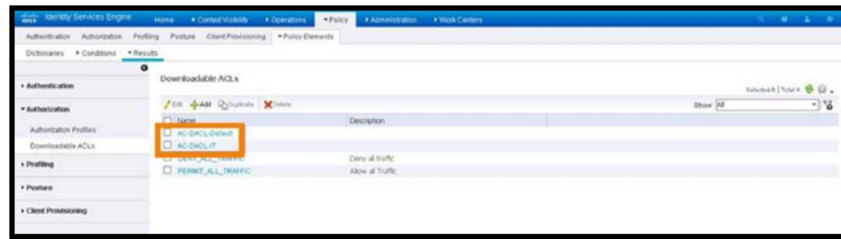
Authorization profiles

1. **Navigate to Policy > Policy Elements > Results > Authorization > Authorization Profiles.** The first two profiles were created for this lab: AC-Auth-Default and AC-Auth-IT.
2. If you drill down into **AC-Auth-Default**, you will see that it references the **DACL AC-DACL-Default**, described below.
3. If you drill down into **AC-Auth-IT**, you will see that it references the **DACL AC-DACL-IT**, described below. It also has two advanced attributes: one for the address pool, and one for the group policy.

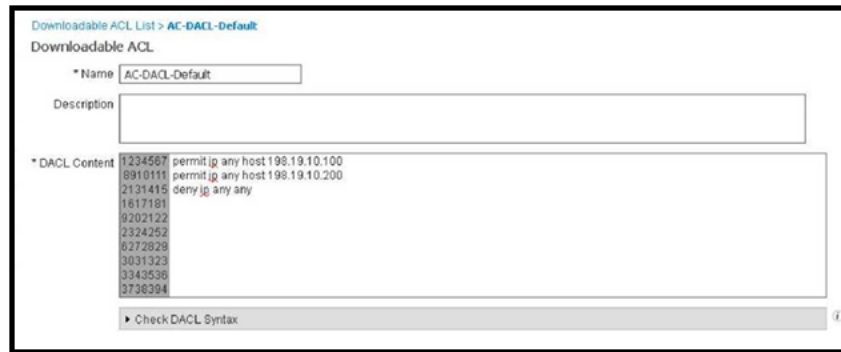


Downloadable ACLs

1. Navigate to **Policy > Policy Elements > Authorization > Downloadable ACLs**. The first two DACLs were created for this lab: **AC-DACL-Default** and AC-DACL-IT.



2. If you drill down into **AC-DACL-Default**, you will see that it restricts access to 198.19.10.100 and 198.19.10.200.



3. If you drill down into **AC-DACL-IT**, you will see that there are no restrictions.



